

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

**An Online Ordering System for Brand-New and Preloved Clothes with
Product Listing, Checkout, and Buyer-Seller Management.**

A Capstone Project Proposal

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Submitted by:

Fernandez Rodalen D.

Naral Mary Kate

Galitguit Andrea

Atis Carl Vincent

Navarra Norman

Second Semester

Academic Year 2026-2027

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The growth of internet technology has greatly changed the way people buy and sell products in today's market. Many businesses now rely on online platforms because they offer faster communication, wider market reach, and more convenient transaction processes. In the Philippines, this shift is especially visible among small businesses that use social media and web-based platforms to promote their products and interact with customers. Clothing businesses, in particular, have benefited from this trend because customers often prefer to browse fashion items online before deciding to purchase them. As a result, an online ordering system has become an important tool for businesses that want to improve their operations and remain competitive in the digital age.

The clothing industry is one of the most active sectors in online selling because fashion products appeal to different types of customers with different needs and preferences. Some buyers look for brand-new clothes because they want new styles and guaranteed product quality, while others choose preloved clothes because they are more affordable and can still be used for a long time. In some businesses, both brand-new and preloved clothes are sold together, which creates a wider range of choices for customers. However, managing these products manually can be difficult, especially when the seller depends only on social media posts and private messages to handle orders. This kind of setup may lead to confusion, delayed responses, incomplete records, and difficulty in monitoring inventory.

At present, many small clothing businesses use Facebook or similar platforms to post product pictures and receive customer inquiries. While this method is

convenient at first, it can become inefficient as the number of customers and items increases. Orders sent through messages may be difficult to organize, and the seller may struggle to keep track of which products are still available, reserved, or already sold. In addition, when communication happens only through chat, there is a higher chance of missed details, repeated questions, and delayed confirmations. These problems show the need for a more organized and reliable system that can handle both product display and order processing in one platform.

An online ordering system can help solve these problems by giving customers a more structured way to browse products, select items, and place orders. For a business that sells both brand-new and preloved clothes, the system can separate items into categories while still keeping everything in one website. This makes it easier for customers to find what they need and compare available options. It also helps the seller manage product information more clearly, including the item's name, price, condition, size, and availability. With a proper system in place, the whole ordering process becomes faster, more accurate, and more convenient for both buyers and sellers.

The inclusion of product listing features is important because clothing items need clear and complete details before customers make a purchase. Buyers usually want to see photos, descriptions, prices, and item conditions before deciding whether to order. This is especially true for preloved clothes, where the condition of the item is a major factor in the buying decision. A well-designed product listing can help the seller present items in a more organized and professional manner. It can also reduce the need for repeated inquiries because customers can already view the important details directly on the website.

Another important part of the system is the checkout process. Checkout allows customers to review the items they selected, confirm their order, and submit the necessary information before completing the transaction. This step is important because it helps prevent mistakes and ensures that all order details are properly recorded. For the seller, checkout makes it easier to track orders and monitor sales activity. It also helps maintain more accurate inventory records, which is very useful for a business that handles both new and preloved items. By automating this part of the process, the business can reduce manual work and improve efficiency.

Buyer-seller management is also a necessary feature in an online clothing system. Customers often need to ask questions about sizes, product condition, shipping methods, payment options, or availability before making a purchase. In the case of preloved clothing, these concerns are even more important because buyers want to make sure that the item still meets their expectations. A system that supports proper buyer-seller communication can make transactions smoother and help build customer trust. It also allows the seller to respond more professionally and manage customer concerns in an organized way. This can lead to better customer satisfaction and stronger relationships between the business and its buyers.

Overall, the proposed **Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management** aims to improve the current way clothing businesses handle online sales. Instead of relying only on manual chatting and informal posting, the business can use a centralized system that organizes products, orders, and communication in one place. This will help reduce errors, save time, and make the selling process more efficient. At the same time, it will provide customers with a better shopping experience because they can easily browse products, place orders, and communicate with the seller in a more organized manner. In this way, the system supports both business growth and customer convenience in a modern online setting.

1.2 Statement of the Problem

This study aims to develop an **Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management**. The growing use of online platforms in business has made it easier for sellers to reach customers, but it has also created challenges in handling orders, organizing product information, and maintaining clear communication between buyers and sellers. In the case of businesses that sell

both brand-new and preloved clothes, these challenges become even more noticeable because each item must be properly described, categorized, and monitored. Since many small clothing businesses still rely on manual processes or social media messaging to manage sales, problems such as delayed responses, inaccurate records, missing order details, and confusion in inventory often occur. Because of these issues, there is a need for a more organized and reliable system that can support the entire ordering process.

The proposed study seeks to improve the current process by developing a web-based system that can manage product listings, order checkout, and buyer-seller communication in one platform. Instead of depending only on private messages, informal posting, and manual tracking, the system will allow customers to browse available clothes, view item details, and place orders through a structured interface. It will also help the seller manage product availability and customer transactions more efficiently. By using a centralized system, the business can reduce common errors and provide a better shopping experience for customers. This study is important because it addresses both the operational needs of the seller and the convenience of the buyer.

Specifically, this study seeks to answer the following questions:

1. What difficulties are experienced by the business in managing brand-new and preloved clothes through manual methods and social media-based selling?
2. How can the proposed system improve the product listing process by organizing brand-new and preloved clothes in a clear and accessible way?
3. How can the checkout feature of the system help make the ordering process more efficient, accurate, and convenient for customers?
4. How can the buyer-seller management feature improve communication, transaction handling, and order coordination?
5. How can the system help the seller monitor product availability and reduce errors in inventory tracking and order processing?
6. How can the proposed system provide a more organized, user-friendly, and reliable online shopping experience for both buyers and sellers?
7. How can the system support the overall business operations of clothing sellers who offer both new and preloved items in one platform?

This study focuses on identifying how technology can be used to improve the way clothing businesses handle online transactions. It also aims to show how a

properly designed ordering system can help reduce manual work, improve customer service, and create a more efficient business process. In doing so, the proposed system is expected to benefit both the business owner and the customers by making online shopping easier, faster, and more organized.

1.3.2 Specific Objectives

General Objective

The general objective of this study is to develop a **web-based ordering system for brand-new and preloved clothes** that can efficiently manage product listing, order processing, checkout, and buyer-seller communication in one centralized platform.

Specific Objectives

Specifically, this study aims to:

1. Design a web-based system that allows the seller to upload, organize, and manage listings for brand-new and preloved clothing items.
2. Develop a checkout feature that enables customers to place orders easily and review their selected items before confirming their purchase.
3. Create a buyer-seller management module that supports communication between customers and the seller regarding product inquiries, order updates, and transaction concerns.
4. Implement an inventory management function that helps the seller monitor available stock and update product status in real time.

5. Improve the accuracy and organization of order processing by reducing manual encoding and relying on a centralized digital platform.
6. Provide a user-friendly interface that makes it easier for customers to browse products, place orders, and complete transactions.
7. Ensure that the system supports both brand-new and preloved clothing items in a single online platform.

1.4 Scope and Delimitation

This study focuses on the development of an Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management for Margaux Collections. The system will cover the main operations of the business, particularly product listing, order placement, inventory tracking, payment handling, queuing, and analytics. It is designed to help the owner manage clothing items more efficiently and to provide customers with a more organized and convenient way of browsing and ordering products. The study will specifically cater to the needs of Margaux Collections, which sells both brand-new items such as bikinis, coordinated sets, and accessories, as well as secondhand items such as pants, skirts, and shorts.

The system will be developed for use by the owner and her customers in the Philippines. It will use web-based technologies such as PHP, MySQL, HTML, CSS, and JavaScript to support the system's functionality. The platform will distinguish between brand-new and preloved clothing items so that products can be categorized properly and displayed in a clear manner. This separation will help both the seller and the buyers easily identify the type and condition of the items available for purchase.

The scope of the study includes the implementation of features that support product browsing, checkout, buyer-seller communication, order management, and inventory monitoring. It also includes queue management for pick-up orders and analytics to help the owner make better decisions based on sales and order data. These features are intended to improve the overall flow of business

operations and reduce the problems caused by manual and social media-based transactions.

On the other hand, this study has several delimitations. The system will not include integration with external logistics providers or third-party delivery tracking services. It will also not have advanced features such as automated size recommendations, as the focus of the study is on basic ordering and management functions. In addition, the project will prioritize web accessibility rather than the development of a mobile application. Since the business is operated solely by the owner, the system will be designed for a single-user business setup without staff handling transactions, inventory, or order processing.

The existing payment methods of Margaux Collections will remain unchanged. GCash will continue to be used for drop-off transactions, cash on delivery (COD) will be used for delivery orders, and in-person payment will be accepted for store pickup. These payment methods will stay the same because they are already familiar to the owner and convenient for the customers. By maintaining the current payment setup, the system can be easier to adopt while still improving the organization and efficiency of the business.

1.5 Significance of the Study

This study is significant because it develops an **Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management** that can improve the way Margaux Collections operates and serves its customers. The proposed system is designed to benefit several stakeholders by making the ordering process more organized, efficient, and convenient. Since the business is currently managed entirely by the owner without any staff, the system is expected to reduce the amount of manual work involved in handling orders, confirming payments, updating inventory, and responding to customer inquiries. Through automation and centralized management, the business can become more efficient and easier to handle on a daily basis.

For the **Margaux Collections owner**, the system provides a practical solution to the challenges of managing a small business alone. At present, the owner manually handles customer orders through Facebook Messenger, checks payments through GCash, and keeps track of product availability on her own. This process can be time-consuming and may lead to errors, delays, or confusion, especially when multiple customers are inquiring or ordering at the same time. With the proposed system, order processing and inventory management can be automated, allowing the owner to focus more on business operations and customer satisfaction. The system also provides sales insights and analytics that can help the owner make informed decisions regarding product restocking, especially in determining which items, such as brand-new bikinis and coordinated sets or preloved pants, skirts, and shorts, are in higher demand. In this way, the system supports not only daily operations but also the long-term growth of the business.

For the **customers**, the system offers a more convenient and organized way of shopping. Instead of depending only on Facebook Messenger conversations, customers can use a dedicated web platform to browse products, view item details, and place orders more easily. This makes the shopping experience more efficient because customers no longer need to wait for repeated back-and-forth messages just to ask about product availability or order status. The system also provides a clearer payment workflow, since customers can follow the existing payment methods used by the business, such as GCash for drop-off orders, cash on delivery (COD) for delivery orders, and in-person payment for store pickup. In addition, the system can help customers track their orders and receive queue numbers for pickup, which improves transparency and reduces confusion. Overall, the proposed system gives customers a smoother and more reliable online shopping experience.

For **future researchers and students**, this study serves as a useful reference for understanding how web-based systems can support small business operations, especially in businesses that sell both brand-new and preloved fashion items. It can be used as a case study in web development, database management, and systems analysis because it combines practical business functions with technical implementation. The study may also help future researchers who are interested in developing systems for micro enterprises, online clothing stores, or businesses that rely heavily on manual transaction handling. In addition, it provides ideas on how a single-owner business can

transition from social media-based selling to a more structured digital platform. As a result, the study contributes not only to the improvement of Margaux Collections but also to the broader body of knowledge in information technology and business system development.

Overall, the significance of this study lies in its ability to improve business efficiency, reduce manual workload, enhance customer convenience, and provide academic value for future researchers. By creating a centralized system for managing brand-new and preloved clothes, the study addresses the actual needs of a small business while also showing how technology can support growth and better decision-making. It demonstrates that even a small, owner-managed business can benefit from a well-designed online ordering system that is tailored to its specific operations and payment practices.

1.6 Definition of Terms

Margaux Collections Ordering System – A web-based platform designed for Margaux Collections that allows product browsing, ordering, queue management, and sales analytics. The system is intended to support the business by organizing both brand-new and preloved clothing items in one centralized platform.

Brand-new – Items that are new, unused, and sold in their original condition. In this study, brand-new items include bikinis, coordinated sets, and accessories.

Preloved – Apparel that has been previously owned but is still in good wearable condition. In this study, preloved items include pants, skirts, and shorts.

Queue Number – An automated number assigned to each order to help organize processing and pickup. It is used to keep transactions in proper order, especially for drop-off locations and store pickups.

Middleware – PHP-based components that manage communication between the frontend and backend of the system. These components handle authentication, sessions, authorization, and data validation to ensure that the system operates properly and securely.

API – PHP interfaces that support system functions such as cart operations, order updates, payment status processing, and real-time dashboard information. These interfaces allow different parts of the system to exchange data efficiently.

Database Gateway – The MySQL layer responsible for storing and retrieving data related to users, products, orders, payments, and transactions. It serves as the connection between the system's interface and its database records.

GCash – A digital wallet used for advance payment in drop-off orders. Customers pay through GCash before the owner delivers the order to the designated drop-off store.

Cash on Delivery (COD) – A payment method in which customers pay in cash when their order is delivered. This method is used for delivery transactions in the business.

Drop-off Location – A specific store or address where the owner delivers orders for customers to pick up after payment through GCash.

Facebook Posts – Product listings published by the owner on her Facebook page, which are currently used as the main way of displaying available items to customers.

Messenger – The Facebook Messenger platform used by customers to communicate with the owner, ask about products, and place orders.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Literature

The Mesh: Why the Future of Business Is Sharing Gansky, 2010

Gansky (2010) discussed the concept of the mesh as a business model in which sharing, access, and connection become more important than ownership alone. Her work explains how modern business practices are changing as consumers become more open to using products through access-based systems rather than depending only on direct ownership. This idea is important in the context of fashion because it shows that products can continue to have value even after they have been used once. Instead of being treated as disposable, items can move through different users and still remain useful and desirable. This perspective is highly relevant to preloved clothing because it supports the idea that second-hand garments are still part of a valuable product cycle.

The study also emphasizes that sharing-based business models depend on trust, convenience, and organized access. Consumers need to feel that the platform is reliable and easy to use before they will participate in shared or reused product

systems. In this sense, the mesh is not only about product circulation but also about the structure that makes circulation possible. A well-organized digital platform helps consumers find items, understand their value, and complete transactions with confidence. This makes the idea useful for understanding how an online shopping environment can encourage customers to consider preloved clothes as practical and appealing choices.

For Margaux Collections, this study is relevant because the business sells both brand-new and preloved clothing items. The mesh concept helps support the inclusion of second-hand products in the same ordering system as new products because it treats both as part of a continuing value chain. It also suggests that digital platforms can help sellers organize products in ways that make them easier to access and purchase. In a small fashion business, this means that the system should not only display products but also help manage how they move from seller to customer. This supports a more flexible and sustainable approach to clothing retail.

Overall, Gansky's discussion provides a useful theoretical basis for the Margaux Collections Ordering System because it shows that clothing can continue to generate value through reuse, circulation, and digital access. The study supports the idea that preloved fashion should be presented as a legitimate and useful retail category rather than as a lesser option. It also reinforces the importance of having a structured online system that makes product access, ordering, and resale more efficient. In this way, the literature helps justify the development of a platform that supports both brand-new and preloved clothing in a shared digital marketplace.

The Economics of Privacy **Acquisti, Taylor, and Wagman, 2016**

Acquisti, Taylor, and Wagman (2016) examined the economics of privacy and explained how people often share personal information online without fully understanding the possible risks involved. Their study is important because

privacy has become a major concern in digital transactions, especially in systems that require users to create accounts, provide contact information, and complete purchases online. The authors show that users do not always make privacy decisions with complete awareness, which makes careful system design very important. In this sense, the study is relevant to any online ordering system that handles personal and transactional data. It helps explain why trust and security are essential in e-commerce platforms.

The study further emphasizes that privacy concerns affect user behavior and decision-making. When people believe that their information may be exposed or misused, they are more likely to hesitate before registering or making a transaction. This means that privacy is not only a legal or technical issue but also a customer experience issue. In online retail, users want reassurance that their details are protected and that the system is reliable. The study therefore shows that protecting privacy is closely connected to encouraging user participation and maintaining confidence in the platform.

For Margaux Collections, this literature is highly relevant because the Ordering System will likely require customers to enter names, contact numbers, addresses, and other transaction-related details. Since the business will operate online, it must make sure that customer information is stored and handled responsibly. A secure login process, proper account management, and controlled access to customer records are important features that can help build user trust. The study supports the idea that an online clothing platform must protect buyer information just as carefully as it manages product listings and payments. This makes privacy a necessary part of the system's design and operation.

Overall, Acquisti, Taylor, and Wagman (2016) provide a strong basis for understanding why online business systems must address privacy concerns in order to succeed. Their work shows that consumers are more comfortable using digital platforms when they feel their information is safe and their personal data is not being misused. For Margaux Collections, this means that the Ordering System should not only be functional and organized but also secure and trustworthy. By giving attention to privacy, the system can improve customer confidence and encourage more successful online transactions. In this way, the literature supports both the technical and relational needs of the proposed platform.

A Study on Relevance of Second-Hand Clothing Retailing in Ethiopia Khurana and Tadesse, 2019

Khurana and Tadesse (2019) studied the relevance of second-hand clothing retailing in Ethiopia and showed that used clothing continues to play an important role in consumer markets. Their study is useful because it confirms that second-hand fashion is not simply an informal or marginal practice, but a legitimate retail activity with continuing consumer demand. The authors highlighted that second-hand clothing remains relevant due to its affordability, availability, and usefulness for a wide range of buyers. This makes the study important for understanding how preloved fashion can remain a practical and valuable part of modern retail. In this sense, second-hand clothing should be viewed as a meaningful product category rather than a lesser alternative to brand-new items.

The study also suggests that second-hand clothing retailing supports consumer access to fashion at lower prices while also promoting reuse. This is important because many consumers choose used clothing not only because it is cheaper, but also because it offers them access to styles they may not otherwise afford. The authors' discussion shows that second-hand clothing can serve both economic and social needs by making fashion more accessible to different market groups. It also reinforces the idea that preloved garments still have worth when they are presented in a way that customers can understand and trust. This makes clear product information and organized presentation especially important in the online context.

For Margaux Collections, this study is directly relevant because the business includes preloved clothing as part of its product offerings. The findings support the idea that second-hand items should be given proper attention in the Ordering System and should not be treated as secondary products. Instead, they should be displayed clearly, categorized properly, and priced in a way that reflects their condition and value. The system can help buyers see preloved clothes as worthwhile purchases by providing enough product detail and making browsing easier. This is especially useful for a clothing business that wants to present both brand-new and preloved items in one organized platform.

Overall, Khurana and Tadesse (2019) provide strong support for the Margaux Collections Ordering System because they show that second-hand clothing retailing has real market value and consumer relevance. Their study helps justify the inclusion of preloved items in a professional online system and supports the idea that resale can be both practical and sustainable. For Margaux Collections, this means the ordering system should help customers browse, compare, and purchase preloved items with confidence. The literature therefore reinforces the importance of structured presentation, accessibility, and value communication in the proposed platform.

Fair Data Is the New Black **Vänskä et al., 2024**

Vänskä et al. (2024) explored how online shopping, data leaks, and sustainable fashion are connected through the idea of fair data. Their study is important because it shows that digital commerce is not only about buying and selling products, but also about how responsibly a platform handles user information. The authors connect sustainability with trust, suggesting that ethical fashion consumption now includes concern for privacy and digital responsibility. This is highly relevant in an era where consumers are increasingly aware of how their personal information is collected and used. The study therefore broadens the understanding of sustainable fashion beyond clothing alone and into the digital systems that support it.

The study also implies that a sustainable fashion platform must be transparent and reliable in its handling of both products and data. If customers feel that a website is careless with their information, they may become less willing to shop online, even if they support the business's products. This means that user trust is part of sustainability because it affects whether consumers feel comfortable participating in digital fashion commerce. The idea of fair data also suggests that businesses should be honest and responsible in the way they present products and interact with buyers. As a result, system design becomes closely tied to ethical business practice.

For Margaux Collections, this study is useful because the business sells both brand-new and preloved clothing through an online platform that will likely collect customer information during ordering and checkout. The study supports the need for a secure and well-organized system that protects users while also presenting products clearly and professionally. Since preloved clothing is often associated with sustainability, the platform should reflect that same responsibility in how it handles data and transactions. This means the ordering system should not only support buying and selling but also create an environment that feels safe and ethically responsible for customers. In this way, the literature reinforces the importance of combining sustainability with digital trust.

Overall, Vänskä et al. (2024) provide a useful perspective for the Margaux Collections Ordering System because they connect online shopping behavior with privacy, data protection, and sustainable fashion values. Their study supports the idea that a fashion platform should be both environmentally conscious and digitally trustworthy. For Margaux Collections, this means the system should help users browse, order, and communicate with confidence while also maintaining responsible handling of customer data. The literature therefore strengthens the argument that modern fashion retail must address both sustainability and privacy in its design. This makes the study highly relevant to the proposed online ordering system.

Consumption Corridors in Fashion

Vladimirova, 2021

Vladimirova (2021) discussed consumption corridors in fashion and examined the idea of upper consumption limits in minimalist fashion challenges. Her study is relevant because it questions excessive consumption and encourages more intentional buying behavior. The author argues that consumers do not need to completely reject fashion in order to be responsible; instead, they can make more mindful choices about what they purchase and how often they buy. This makes the study important for understanding the role of fashion in sustainable living. In this way, the literature presents fashion consumption as something that can be managed more responsibly rather than simply expanded without limit.

The study further explains that consumption corridors help people balance personal style with environmental responsibility. Rather than encouraging total abstinence from fashion, the concept supports measured and thoughtful purchasing that respects both consumer needs and ecological concerns. This is important because it shows that consumers can still express themselves through clothing while avoiding wasteful habits. It also reinforces the idea that preloved fashion can serve as a practical alternative to buying new items all the time. By choosing second-hand clothing, consumers participate in a more sustainable and less wasteful form of fashion consumption.

For Margaux Collections, this study is useful because it supports the inclusion of preloved clothing as part of a responsible retail model. The business can use its Ordering System to present both brand-new and preloved items in a way that encourages thoughtful purchasing. When product details, item conditions, and prices are clearly displayed, customers can make informed decisions that align with both style and sustainability. The system can therefore help buyers see preloved clothes as valuable options rather than temporary or low-status alternatives. This makes the study relevant to both product presentation and sustainable retail practice.

Overall, Vladimirova (2021) provides a strong theoretical basis for the Margaux Collections Ordering System because it connects fashion consumption with sustainability, restraint, and informed decision-making. The study supports the idea that clothing retail can promote more responsible consumer behavior when products are presented clearly and thoughtfully. For Margaux Collections, this means the ordering system should help customers browse and choose items in a way that supports both convenience and conscious consumption. The literature therefore strengthens the argument that preloved fashion has an important place in modern online retail.

Used-Clothing Donations and Apparel Production in Africa Frazer, 2008

Frazer (2008) examined used-clothing donations and apparel production in Africa and showed how second-hand garments are connected to broader systems of

trade, reuse, and consumption. His study is important because it explains that used clothing is not merely discarded material, but part of an active flow of goods that continues to create value after first use. The author's discussion highlights how donated and reused clothing can affect local markets, consumer choices, and production systems. This makes the study useful for understanding the wider social and economic role of preloved garments. In this sense, second-hand clothing should be seen as a legitimate and meaningful product category within fashion retail.

The study further suggests that used-clothing circulation can create opportunities for both sellers and buyers. Sellers can recover value from items they no longer need, while buyers gain access to affordable clothing options that remain useful and wearable. This cycle of reuse helps reduce waste while also supporting more practical forms of consumption. It shows that clothing can continue to serve a purpose even after it has been passed on from one owner to another. For retail systems that include preloved products, this means that the process of buying and selling second-hand items can contribute to both economic activity and sustainability.

For Margaux Collections, this literature is relevant because the business includes preloved clothes as part of its offerings. The study supports the idea that these items should be organized and presented as valuable products within the Ordering System rather than as leftover stock. A structured online platform can help the business manage listings, track product availability, and present second-hand garments in a more professional way. It can also help customers understand that preloved clothes still have practical and commercial worth. This is important because the business aims to serve buyers who are looking for both affordability and style.

Overall, Frazer (2008) provides useful support for the Margaux Collections Ordering System because it shows that second-hand clothing belongs to a larger system of reuse that benefits both consumers and sellers. The study helps justify the inclusion of preloved garments in an online retail platform and strengthens the argument that resale is a meaningful form of clothing circulation. For Margaux Collections, this means the ordering system should help make second-hand items easy to browse, understand, and purchase. The literature therefore supports the idea that preloved fashion can be both sustainable and commercially valuable when organized properly.

I Rent, Swap or Buy Second-Hand Brand, Jacobs, and Taljaard-Swart, 2023

Brand, Jacobs, and Taljaard-Swart (2023) examined the reasons consumers rent, swap, or buy second-hand clothing online. Their study is relevant because it shows that second-hand clothing consumption is not driven by only one motivation, but by a combination of practical, social, and environmental factors. The authors explain that people engage in collaborative clothing consumption for reasons such as convenience, affordability, sustainability, and variety. This makes the study important for understanding why consumers may choose preloved fashion instead of brand-new clothing. In this sense, second-hand fashion is presented as a flexible and meaningful part of modern consumer behavior.

The study also emphasizes that the online environment plays an important role in shaping consumer participation in collaborative clothing models. Buyers are more likely to engage in renting, swapping, or purchasing second-hand items when the system is easy to use and the product information is clear. This means that platform design influences not only access to products but also customer willingness to participate in second-hand fashion markets. The authors' findings show that digital convenience can help normalize alternative fashion behaviors and make them more attractive to consumers. Clear presentation, reliable communication, and organized product access all contribute to this positive experience.

For Margaux Collections, this study is relevant because it supports the idea that customers may be open to both brand-new and preloved clothing when the system makes product browsing and ordering simple. The Ordering System can help by presenting products clearly and allowing customers to compare items quickly. Since the business offers clothing that serves different customer needs, the platform should support a smooth and trustworthy shopping experience. This study also supports the importance of designing the system in a way that makes

second-hand products feel accessible and legitimate. In this way, the literature aligns with the goal of combining convenience, value, and sustainability in one platform.

Overall, Brand, Jacobs, and Taljaard-Swart (2023) provide useful support for the Margaux Collections Ordering System because they show that consumers are willing to participate in second-hand clothing markets when the process is convenient and understandable. Their study helps explain why an online platform must be user-friendly, organized, and informative in order to encourage purchases. For Margaux Collections, this means that the ordering system should not only manage transactions but also make preloved clothing feel practical and appealing. The literature therefore supports the development of a digital clothing platform that is efficient, accessible, and responsive to customer behavior.

The International Trade in Secondhand Clothing Abimbola, 2012

Abimbola (2012) studied the international trade in second-hand clothing and focused on the problem of information asymmetry between West African and British traders. The study is important because it shows that the second-hand clothing market depends heavily on the quality, accuracy, and completeness of information shared between buyers and sellers. When one side knows more than the other, trust becomes harder to establish and transactions become more difficult to manage. This makes the study highly relevant to online clothing retail, where customers often cannot inspect items physically before purchasing. In this sense, transparency becomes a major factor in second-hand fashion markets.

The study further explains that information asymmetry creates uncertainty in resale markets, especially when product details are incomplete or unclear. Buyers want to know exactly what they are purchasing, while sellers need to present items in ways that make them appear trustworthy and worthwhile. The author's discussion shows that second-hand clothing works best when product information is shared clearly and consistently. This includes accurate

descriptions, item condition, and other details that reduce doubt. When information is incomplete, buyers may hesitate or avoid completing the transaction.

For Margaux Collections, this study is directly relevant because the business sells preloved clothing that customers will only be able to evaluate through the online ordering system. Since buyers cannot physically inspect the items, the system must provide enough information to reduce uncertainty and build trust. Detailed product listings, clear condition labels, and organized categories can help make preloved clothes more appealing and dependable in the eyes of the customer. The Ordering System can therefore serve as a tool for improving transparency and reducing the gap between seller knowledge and buyer expectations. This is especially important for maintaining confidence in the purchasing process.

Overall, Abimbola (2012) provides strong support for the Margaux Collections Ordering System because it shows that second-hand clothing retail depends heavily on clear communication and reliable product information. The study helps justify the need for an organized digital platform where customers can understand what they are buying before they place an order. For Margaux Collections, this means the system should be designed to reduce uncertainty and support fair, informed transactions. The literature therefore reinforces the importance of transparency, trust, and product clarity in online preloved fashion retail.

The Work of Shopping: Resellers and the Informal Economy at the Goodwill Bins

Ayres, 2019

Ayres (2019) explored the work of shopping and the role of resellers in the informal economy at the goodwill bins. Her study is relevant because it shows that resale activity involves more than simply buying and selling old items; it also requires judgment, effort, and knowledge about product quality and value. The

author presents reselling as a form of work that involves careful selection, evaluation, and presentation of goods. This makes the study useful for understanding second-hand fashion as a serious economic activity rather than an informal leftover market. In this sense, the resale process becomes part of a broader business practice that generates value for both sellers and buyers.

The study further explains that resellers must recognize which items are worth keeping, pricing, and offering to customers. This means that second-hand retail depends on skills such as product assessment, trend awareness, and customer understanding. Ayres shows that the process of finding valuable used items takes time and effort, which makes resale an active and meaningful part of the informal economy. It also suggests that the way products are selected and displayed can strongly affect whether buyers will be interested. This is important because second-hand fashion is not only about availability but also about how items are curated and presented.

For Margaux Collections, this study is relevant because the business also deals with preloved clothing that must be carefully organized and presented online. The Ordering System can support this by making it easier to list products, track inventory, and present items in a more professional way. Since the owner must manage both brand-new and preloved items, the system can help reduce manual work and make the resale process more structured. It also allows the business to reflect the effort involved in selecting and offering second-hand items. In this way, the study supports the value of having a system that helps turn resale activity into a more organized business operation.

Overall, Ayres (2019) provides useful support for the Margaux Collections Ordering System because it shows that second-hand retail depends on careful product selection, organization, and seller judgment. The study helps explain why preloved items should be managed in a structured way rather than treated casually. For Margaux Collections, this means the ordering system should help the owner organize products clearly and present them in a way that builds customer confidence. The literature therefore reinforces the importance of professionalism, curation, and efficient management in an online preloved clothing platform.

Overdressed: The Shockingly High Cost of Cheap Fashion Cline, 2013

Cline (2013) examined the hidden costs of cheap fashion and argued that low-priced clothing often carries social and environmental consequences that are not immediately visible to consumers. Her work is important because it challenges the assumption that inexpensive clothing is always the best or most practical choice. Instead, the author shows that fast fashion can encourage waste, overconsumption, and a reduced appreciation for the true value of clothing. This makes the study highly relevant to discussions about sustainable fashion and second-hand retail. In this sense, clothing should be understood not only as a consumer product but also as part of a larger system of responsibility and impact.

The study further explains that cheap fashion often leads to faster turnover of garments, which contributes to textile waste and weakens the long-term value of clothing. Cline's discussion encourages consumers to think more carefully about what they buy and how often they replace items. This is important because it shows that fashion decisions are tied to sustainability, quality, and long-term use. When consumers choose cheaper products without considering these factors, they may end up supporting practices that are harmful to both society and the environment. The book therefore provides a strong argument for more thoughtful and responsible clothing consumption.

For Margaux Collections, this study is relevant because the business includes preloved clothing in its product offerings, giving customers an alternative to fast fashion. The Ordering System can help present preloved items as practical, stylish, and responsible choices rather than as lower-quality substitutes. By showing product details, item condition, and price clearly, the system can help customers recognize the value of second-hand clothing. This supports the idea that preloved fashion can offer affordability without encouraging wasteful buying habits. In this way, the study aligns closely with the goal of promoting both business convenience and more sustainable consumer behavior.

Overall, Cline (2013) provides strong support for the Margaux Collections Ordering System because it highlights the broader consequences of cheap fashion and the importance of valuing clothing more carefully. The study helps justify the inclusion of preloved items as part of a more responsible retail model. For Margaux Collections, this means the system should not only facilitate sales but also encourage customers to view clothing as something with lasting value. The literature therefore reinforces the importance of sustainability, product value, and mindful consumption in the proposed online ordering platform.

“Mix-and-Match” Fashion Trend and Luxury Brand Recognition Amatulli et al., 2016

Amatulli et al. (2016) investigated the relationship between fashion trends and luxury brand recognition using eye-tracking methods. Their study is relevant because it shows that visual presentation strongly influences how consumers notice, evaluate, and remember fashion items. The authors demonstrate that consumers often respond first to what they see before they fully assess product value or quality. This makes visual design an important factor in fashion retail, especially in online shopping where customers rely heavily on images. In this sense, the way a product is displayed can affect whether it attracts attention and creates a positive impression.

The study further suggests that visual cues play a major role in shaping consumer interest and brand perception. When items are arranged clearly and attractively, consumers are more likely to focus on them and associate them with quality or desirability. This is especially important in fashion because product

appearance is often tied to style, identity, and personal preference. Amatulli et al. show that visual recognition is not accidental but influenced by how the product is framed in the consumer's view. This finding is useful for understanding how product display can affect shopping behavior and decision-making.

For Margaux Collections, this study is highly relevant because the business will present clothing items through an online ordering system where customers cannot physically examine the products. This means that images, layout, and display quality become essential in helping buyers evaluate both brand-new and preloved items. The Ordering System should therefore present products in a clean, attractive, and consistent manner so that customers can quickly understand what is being offered. Good presentation can make preloved items look more desirable and professional, which may increase customer confidence and interest. In this way, the study supports the importance of product visualization in online fashion sales.

Overall, Amatulli et al. (2016) provide useful support for the Margaux Collections Ordering System because they show that visual design strongly affects consumer attention and product perception. The study helps explain why online clothing platforms must focus not only on product listing but also on how items are shown to buyers. For Margaux Collections, this means the system should make clothing look organized, appealing, and easy to evaluate. The literature therefore reinforces the importance of product presentation, visual clarity, and consumer perception in the success of the proposed platform.

Foreign Studies

Foreign studies provide empirical evidence that supports the design and purpose of the proposed system. These studies are especially valuable because they show how consumers and businesses actually behave in real settings, which helps justify the need for features such as product listings, inventory control, transparent information, and digital trust-building. In the context of Margaux Collections, the foreign studies reviewed here explain why product presentation, consumer confidence, sustainability, and system organization are essential in a digital fashion retail environment. They also demonstrate that second-hand fashion is influenced by emotional, practical, and environmental considerations that must be reflected in the design of the ordering system.

Market Prices of Remanufactured, Used, and New Items Neto, Bloemhof, and Corbett, 2016

Neto, Bloemhof, and Corbett (2016) conducted a study that examined the market prices of remanufactured, used, and new items using eBay data. Their research focused on how consumers value products in different conditions and how pricing reflects market demand. The study is useful because it shows that used products have their own market value and are not automatically considered inferior to new

items. This is relevant to Margaux Collections because it supports the idea that preloved items should be priced and presented in a way that reflects their actual worth.

The study found that used items usually fall between new and remanufactured products in terms of price. This suggests that buyers evaluate second-hand goods by balancing affordability and condition rather than by price alone. For Margaux Collections, this means that preloved items should be priced transparently and fairly based on their condition, appeal, and remaining value. Clear pricing can help customers see preloved products as practical and worthwhile purchases instead of simply cheaper alternatives.

Neto, Bloemhof, and Corbett (2016) also showed that product presentation affects how consumers judge value. When descriptions are clear and pricing is consistent, buyers are more likely to trust the seller and consider the product a good deal. This supports the need for detailed product information, condition labels, and organized listings in the Margaux Collections Ordering System. These features can help reduce hesitation and make customers more confident in their purchase decisions.

The study further implies that transparent pricing practices can strengthen customer trust and support repeat purchases. When consumers understand how an item is valued, they are more likely to feel comfortable buying from the same seller again. For Margaux Collections, this means that the ordering system should display prices clearly and consistently so that buyers can quickly understand the value of each item. In this way, the system supports both professionalism and customer confidence.

Consumers' Value and Risk Perceptions of Circular Fashion Kim, Jung, and Lee, 2021

Kim, Jung, and Lee (2021) conducted a quantitative study to examine consumers' value and risk perceptions toward circular fashion. Their research focused on how factors such as perceived quality, emotional value, social value, environmental value, economic value, and perceived risk influence consumers'

intention to purchase circular fashion products. The study is useful because it shows that second-hand and preloved fashion are not evaluated only by price, but also by the meanings and benefits consumers associate with them. This makes the study relevant to Margaux Collections because it supports the idea that preloved items can be presented as valuable fashion products rather than merely used goods.

The study found that perceived quality, emotional value, social value, and environmental value significantly affect purchase intention. This means that consumers are more likely to buy circular fashion products when they believe the items are stylish, meaningful, and aligned with sustainable consumption. The findings are important for Margaux Collections because they suggest that preloved items should be displayed in a way that enhances their attractiveness and desirability. If the ordering system presents products professionally, customers may see them as worthwhile and fashionable purchases rather than second-hand alternatives.

Kim, Jung, and Lee (2021) also emphasized the role of trust and complete product information in reducing consumer hesitation. Since buyers cannot inspect preloved items physically, they depend heavily on photographs, descriptions, and condition details before deciding to purchase. This supports the need for clear product images, accurate descriptions, and condition labels in the Margaux Collections Ordering System. These features can help reduce uncertainty and improve customer confidence. In online fashion retail, accurate information is essential because it shapes how buyers evaluate both the quality and value of an item.

The study further implies that circular fashion can succeed when businesses present products in a professional and organized way. Consumers are more likely to respond positively when they can easily understand what the item looks like, what condition it is in, and why it is valuable. For Margaux Collections, this means that the system should not only function as a sales platform but also as a presentation tool for preloved fashion. By combining clear visuals, organized listings, and informative product details, the ordering system can help make preloved items more appealing and trustworthy to customers.

Consumers' Perceptions and Purchasing Behavior for Second-Hand Luxury Goods

Tu, Hsu, and Creativani, 2022

Tu, Hsu, and Creativani (2022) conducted a quantitative study in the United States to examine consumer behavior toward second-hand luxury goods. The researchers used survey data and structural equation modeling to analyze how perceived quality, emotional value, social value, environmental value, economic value, and perceived risk influence purchase intention. Their findings are useful because they show that second-hand fashion purchases are shaped by both rational and emotional factors. This is significant because it means that consumers do not evaluate preloved products only on the basis of price. Instead, they also consider how the item makes them feel, how it reflects their identity, and whether it aligns with their values.

The study found that perceived quality, emotional value, social value, and environmental value were the strongest predictors of purchase intention. This means that consumers are more likely to buy second-hand luxury items when they believe the products are stylish, meaningful, and environmentally responsible. Price matters, but it is not the only factor. This has direct relevance to Margaux Collections because it suggests that preloved items should be presented in a way that increases their aesthetic and emotional appeal. If items are displayed professionally and described in a compelling way, they are more likely to be seen as desirable rather than merely used or discounted products. This reinforces the importance of thoughtful product presentation in the proposed ordering system.

The researchers also emphasized the importance of trust and complete product information. Since buyers cannot inspect the item physically, they depend on the seller's descriptions and photographs. When the information is accurate and

detailed, customers feel more confident about making a purchase. This supports the inclusion of multiple product images, condition labels, and detailed descriptions in the Margaux Collections Ordering System. These features can reduce hesitation and improve customer satisfaction. In online second-hand retail, trust is often built through transparency, and this study shows that businesses must provide enough information for consumers to feel secure in their decisions.

Another important implication of the study is that second-hand luxury goods can carry symbolic value beyond their material condition. Consumers may see them as fashionable, exclusive, or meaningful purchases that reflect a certain lifestyle or value system. This idea is useful for Margaux Collections because it shows that preloved items can be positioned as curated fashion pieces rather than simply second-hand goods. If the system presents the products with consistent styling, clear categories, and attractive visuals, it can help create a stronger sense of value and desirability. The study therefore supports not only the technical design of the system, but also the way products should be marketed within it.

Selling Second-Hand Luxury: Empowerment and Social Roles

Turunen, Cervellon, and Carey, 2020

Turunen, Cervellon, and Carey (2020) conducted a qualitative study that explored the experiences of people who sell second-hand luxury goods. Through interviews, they found that sellers often experience a sense of empowerment because they are able to recover value from unused items and participate in a more sustainable fashion economy. This is important because it shows that selling second-hand products is not only a financial activity but also an identity-forming one. The study highlights the personal and social meaning of resale, suggesting that second-hand selling can influence how individuals view their role in consumer culture and sustainability.

The study found that sellers often view themselves as curators and responsible participants in fashion circulation. They also use storytelling to make items more appealing to buyers. By sharing the background, condition, or style significance of each item, sellers create stronger connections between products and consumers. This is highly relevant to Margaux Collections because it supports the idea that product descriptions should do more than list facts; they should also help build meaning and interest. A well-written product description can increase a buyer's curiosity and help the item feel more valuable, even if it is pre-owned.

The findings also suggest that organized presentation increases professionalism and trust. Sellers who manage their items carefully and communicate clearly are more likely to attract buyer confidence. For Margaux Collections, the Ordering System can support this by helping the owner organize product information in a structured and attractive way. In this way, the system becomes a tool not only for sales but also for brand development. It allows the business to present itself as organized, thoughtful, and professional, which is especially important in a market where customers often make judgments based on the quality of product presentation.

Another useful point from the study is that second-hand selling can strengthen the seller's sense of agency. Instead of viewing resale as a simple disposal activity, sellers may see it as a meaningful business practice that combines economic, social, and environmental goals. This is relevant to small businesses like Margaux Collections because it shows that an ordering system can support not just transactions, but also a broader business identity. By making it easier to manage items, document product details, and communicate with customers, the system can help the owner engage more confidently in the selling process. This contributes to both business efficiency and owner empowerment.

Online Shopping vs. Offline Shopping: A Comparative Study

Sarkar and Das, 2017

Sarkar and Das (2017) conducted a comparative study of online and offline shopping behavior to determine which factors influence consumer preference.

Their research showed that online shopping offers clear benefits, especially in terms of convenience, broader product access, and the ability to compare options quickly. These advantages explain why many customers prefer digital platforms when the shopping experience is simple and trustworthy. The study is useful because it identifies the strengths of online shopping that businesses can maximize when designing a digital ordering system.

The study also found that online shopping has drawbacks, especially when buyers cannot inspect products personally. Consumers may worry about quality, security, and delivery reliability. These concerns are particularly important in fashion retail because clothing fit and appearance are difficult to evaluate from images alone. This supports the need for detailed product presentation and reliable ordering features in Margaux Collections. Customers who cannot examine clothing physically are more likely to depend on the quality of the information they see online, which makes accuracy and clarity essential.

Their findings further show that trust is a major factor in online purchasing decisions. Buyers are more likely to complete a purchase when the platform appears organized, safe, and easy to use. This is why the Margaux Collections Ordering System should prioritize usability, clear navigation, and transparent transaction handling. A professional system can reduce uncertainty and make online shopping feel more dependable. In the context of a small business, this trust is especially valuable because it can help turn first-time buyers into repeat customers.

The study also highlights that convenience alone is not enough to sustain customer interest. Even when a digital platform is accessible, consumers may still hesitate if the system appears confusing or if the seller does not provide enough product information. This is particularly important for a fashion business that sells both brand-new and preloved items. The ordering system must therefore combine convenience with confidence-building features such as item descriptions, product photos, and order tracking. By doing so, Margaux Collections can create a shopping experience that is both efficient and reassuring.

Sustainable Supply Chain Management

Svensson, 2007

Svensson (2007) examined Sustainable Supply Chain Management as a conceptual framework that integrates economic, environmental, and social concerns into business operations. The study is relevant because it explains why sustainability should be built into the structure of a business rather than treated as an afterthought. For Margaux Collections, this means that inventory and ordering systems should help support responsible use of resources and efficient product flow. Sustainability is not presented merely as an ethical ideal, but as a practical business principle that can improve long-term performance.

The study emphasized that sustainability depends on coordination and information sharing. When businesses maintain accurate records and use technology to support operations, they reduce waste and improve decision-making. This is highly applicable to small businesses with limited inventory, where errors can easily lead to overstocking or stock shortages. The Margaux Collections Ordering System addresses this need through real-time product updates and better inventory tracking. By ensuring that records remain current, the system can help the business avoid unnecessary duplication and better align product availability with demand.

Another important contribution of the study is its focus on the role of information systems in supporting sustainability. Digital tools help businesses monitor movement, improve forecasting, and strengthen transparency. These functions are essential for a preloved fashion business because they allow the owner to understand what items are selling, which products remain available, and how the business can manage stock more responsibly. In this sense, the system contributes not only to organization but also to sustainable business practice. It supports a more efficient cycle of buying, selling, and reusing items, which is central to responsible retailing.

The study also shows that sustainability is connected to efficiency and accountability. A business that keeps accurate records is better able to make

informed decisions about stock replacement, product promotion, and waste reduction. This is especially important for a small fashion business that handles unique items and limited stock. For Margaux Collections, the ordering system can help ensure that products are managed responsibly from listing to sale. As a result, sustainability becomes part of everyday operations rather than a separate concern.

Clothing Reuse and Environmental Impact

Cuc and Vidovic, 2014

Cuc and Vidovic (2014) discussed the environmental impact of clothing reuse and recycling, emphasizing that extending the life of garments can significantly reduce waste and resource consumption. Their study is important because it reinforces the sustainability value of second-hand fashion. When garments are reused instead of discarded, fewer resources are needed for new production, and textile waste is reduced. This makes second-hand fashion a practical response to environmental concerns in the clothing industry.

The study showed that even a short extension in garment life can create meaningful environmental benefits. This means that buying and selling preloved clothing is not only economically practical but also environmentally responsible. For Margaux Collections, this supports the idea of presenting preloved items as part of a circular fashion approach. Customers should be made aware that purchasing preloved clothes helps reduce waste and gives garments a second life. The ordering system can help communicate this message by including short sustainability statements in product pages or item descriptions.

The researchers also discussed the importance of consumer education and proper garment care. By giving buyers simple care advice, businesses can help prolong the lifespan of clothing. This is relevant to the Ordering System because product pages can include care tips along with descriptions and images. In this way, the system can promote both sales and responsible use. It becomes not only a transaction platform but also a source of helpful information that encourages customers to care for their purchases properly.

Another important implication of the study is that environmental responsibility can be made more visible through retail systems. When customers understand that their purchase contributes to waste reduction, they may develop a stronger appreciation for preloved products. This can improve the business's image while also encouraging more thoughtful consumption. For Margaux Collections, this means that sustainability can be integrated into both the content and the structure of the ordering system. By doing so, the business can appeal to consumers who value both fashion and environmental responsibility.

Closing Connection

Together, these foreign literature and studies show that second-hand fashion, digital ordering systems, and sustainable business practices are closely connected. They support the idea that Margaux Collections needs a system that is not only functional but also customer-oriented, transparent, and professionally structured. The evidence from both local and foreign sources strongly justifies the development of the proposed Ordering System as a practical solution for improving business operations, strengthening trust, and supporting sustainable fashion retail.

Local Literature

The Philippine context provides a strong foundation for understanding why the Margaux Collections Ordering System is both relevant and necessary. In local literature, the discussion of second-hand fashion, digital commerce, microbusiness operations, and government support for digital transformation shows that the proposed system is not only technically useful but also aligned with current social and economic realities in the Philippines. This section strengthens the study by connecting the business needs of Margaux Collections with established local concepts, policies, and industry practices. It also demonstrates that the development of a more structured ordering platform is consistent with the broader direction of Philippine business modernization and consumer behavior.

Investigating Second-Hand Fashion Trade and Consumption in the Philippines

Isla, 2013

The Philippines has long had a strong cultural tradition of second-hand fashion consumption through what is commonly known as *ukay-ukay*. The term comes from the Visayan word for rummaging or searching, and it refers to the practice of buying imported pre-owned clothing that is often sold by weight or at very low prices. Isla (2013), in one of the earliest academic studies on this topic, examined how *ukay-ukay* is connected not only to economic realities but also to Filipino culture, identity, and everyday fashion practices. Drawing on ethnographic research conducted in several Philippine cities, the study traced the flow of second-hand garments, which are commonly sourced from textile donations and surplus clothing from countries such as the United States, Europe, and Japan. More importantly, Isla analyzed how Filipino consumers assign meaning and value to these items through the act of buying, wearing, and sharing them. The study therefore presents *ukay-ukay* not merely as a retail activity, but as a social and cultural practice that reflects the changing lifestyle of Filipino consumers.

One of the key contributions of the study is its argument that *ukay-ukay* is not simply a low-income shopping option. While affordability is certainly one reason people buy second-hand clothing, Isla (2013) showed that the practice has deeper cultural meaning. For many Filipinos, *ukay-ukay* is associated with resourcefulness, creativity, and the excitement of finding valuable pieces at affordable prices. Shoppers often take pride in discovering branded or high-quality items that would normally be out of reach if purchased new. In this way, second-hand fashion becomes part of a broader experience of aspirational consumption, where individuals can enjoy fashionable items without paying full retail prices. The practice is therefore both practical and symbolic, combining economic advantage with personal satisfaction and social meaning. This makes *ukay-ukay* especially important in the Philippines, where budget-conscious

buying is often combined with a strong interest in style, uniqueness, and value for money.

The study also revealed that *ukay-ukay* shopping often has a social dimension. People frequently go shopping together with friends or family, turning the process of searching through piles of clothing into a shared activity with its own cultural enjoyment. This kind of shopping is not always fast or purely functional; instead, it can be a social and even recreational experience. The act of browsing, comparing, and discovering hidden finds gives second-hand shopping a distinct character that many consumers value. Isla (2013) also noted that younger, college-educated Filipinos have increasingly embraced *ukay-ukay* as a form of conscious consumption and fashion expression, especially in urban areas where sustainability and personal style are more visible concerns. This shows that second-hand fashion in the Philippines is not limited to necessity-driven consumers but also appeals to those who value individuality and environmental awareness. It suggests that the market for preloved clothing is not only broad but also evolving, with different consumer segments assigning different meanings to the same product category.

The findings of the study are directly relevant to Margaux Collections, because the business operates in a Philippine context where second-hand fashion is already familiar and widely accepted. However, Margaux Collections occupies a space between traditional *ukay-ukay* and more formal second-hand retail. Traditional *ukay-ukay* is often informal, unbranded, and sold by weight, while more formal second-hand stores usually organize products individually and present them in a more curated manner. The Margaux Collections Ordering System can help the business combine the cultural familiarity of *ukay-ukay* with the structure and professionalism of modern e-commerce. This means that the business can preserve the appeal of second-hand fashion while also offering customers clearer product information, better organization, and a more trustworthy shopping experience. In doing so, the system allows the business to maintain the charm of preloved fashion while improving the overall quality of the buying process.

This is important because many Filipino consumers enjoy the feeling of discovering second-hand items, but they also want convenience and reliability. Traditional *ukay-ukay* shopping can be enjoyable, but it may also involve uncertainty about product quality, pricing, and availability. A digital ordering

system helps reduce these issues by making it easier to browse products, view clear photos, and understand item conditions before purchasing. For Margaux Collections, this means that preloved items can be presented with care, accurate descriptions, and transparent prices. Customers can still experience the excitement of second-hand fashion, but with more confidence and less confusion. In this sense, the system does not replace the cultural value of *ukay-ukay*; rather, it modernizes its presentation and makes it more suitable for online commerce.

Isla (2013) also observed that the digitization of second-hand fashion trade in the Philippines was still limited at the time of the study, with many transactions happening in physical markets or through informal online spaces such as Facebook groups and Messenger chats. Although the online environment has changed significantly since then, the basic need for more formal and structured digital platforms remains relevant. Informal online selling may be convenient at first, but it often lacks the features needed to build strong customer trust, manage orders efficiently, and maintain consistent product information. This is exactly where the Margaux Collections Ordering System becomes valuable. It provides a more organized and dependable alternative to informal selling methods. By doing so, it supports smoother communication, more accurate order handling, and a more professional digital presence for the business.

Another important insight from the study is that second-hand fashion can reflect identity and cultural participation, not just financial necessity. For many Filipino consumers, *ukay-ukay* is part of how they express taste, resourcefulness, and creativity. The same can be true for preloved products sold through Margaux Collections. If the system presents items attractively and professionally, customers may see them not as leftovers or used goods, but as meaningful fashion choices that fit their personal style. This supports the business's ability to position preloved items as desirable, practical, and culturally relevant. It also shows that a well-designed ordering system can influence how customers interpret and value second-hand products, turning them into items of fashion significance rather than simple resale goods.

Overall, Isla's study provides an important local foundation for the proposed system because it explains why second-hand fashion is already familiar in the Philippines and how its value extends beyond price alone. The findings show that Filipino consumers are open to preloved fashion when it is presented in a way

that reflects quality, identity, and practicality. This supports the need for a system like Margaux Collections Ordering System, which can improve how preloved items are organized, displayed, and purchased. In doing so, the system responds not only to business needs but also to local consumer culture and expectations.

Social Media Utilization of Microbusinesses in Northern Philippines

Binuluan et al., 2024

The use of social media as a primary business platform among Filipino microbusinesses has become increasingly significant, especially as digital commerce continues to expand across Southeast Asia. Binuluan, Gonayon, Mabborang, Panaligan, and Ramos (2024) examined how microbusinesses in a component city in Northern Philippines use platforms such as Facebook and Instagram to market their products, communicate with customers, and process orders. Using a mixed-methods approach that combined survey responses with in-depth interviews, the study provided a clearer picture of how small business owners rely on social media to sustain and manage their operations. Their findings show that social media is no longer used only for promotion, but has become a central part of everyday business activity for many micro-entrepreneurs.

One of the main findings of the study is that Facebook remains the most widely used platform for microbusiness commerce in the Philippines. The majority of respondents used Facebook for product promotion, customer inquiries, and order processing. In many cases, business owners also depended on Messenger and Facebook Marketplace to handle day-to-day transactions. Some even combined these platforms with mobile payment applications to complete the sales process. This shows how deeply embedded Facebook has become in the business practices of small Filipino entrepreneurs. For many microbusinesses, it functions as a storefront, communication channel, and sales tool all at once.

However, the study also pointed out several problems with this informal and highly manual approach. One of the most common concerns was the difficulty of tracking orders and payments. Because transactions were often managed

through Messenger conversations, business owners had to manually remember or record customer details, which increased the risk of confusion and error. Another challenge was inventory management. Since Facebook does not provide a built-in system for organizing products in a searchable or structured way, owners often had difficulty keeping track of what items were available, what had already been sold, and which products needed to be restocked. The lack of a permanent product catalog also made it harder for customers to browse items later or return to specific listings. In addition, managing many customer messages individually through Messenger took a great deal of time and effort, especially for owners handling orders on their own.

These findings are very similar to the situation of Margaux Collections, which currently relies on Facebook and Messenger for product promotion, customer communication, order taking, payment confirmation, and delivery coordination. In this sense, the study by Binuluan et al. (2024) accurately reflects the operational realities faced by the business. Like the microbusinesses in the study, Margaux Collections benefits from the reach and familiarity of Facebook, but it also experiences the limitations of a system that is not designed specifically for business operations. This is why the proposed Ordering System is so important. It responds directly to the weaknesses identified in the study by offering a more organized and efficient digital platform for managing the business.

The local literature makes the case even stronger because it shows that the proposed system fits Philippine culture, policy, and business practice. Isla (2013) establishes that second-hand fashion already has deep roots in the Philippines through ukay-ukay, so Margaux Collections is operating in a market where preloved items are already familiar and socially accepted. Binuluan et al. (2024) confirm that Facebook and Messenger dominate microbusiness commerce, but also reveal the limits of informal selling, especially in order tracking, inventory control, and time-consuming communication. The DICT's Philippine Digital Transformation Strategy 2022 adds policy support by encouraging MSMEs to adopt digital tools that improve business efficiency and competitiveness.

Philippine Digital Transformation Strategy 2022

DICT, 2022

The Philippine Digital Transformation Strategy 2022 reflects the government's broader effort to encourage digital adoption among businesses, public institutions, and communities. The Department of Information and Communications Technology emphasized that digital systems are essential for improving productivity, expanding access to markets, and strengthening the competitiveness of micro, small, and medium enterprises. This policy direction is especially important in a country where many small businesses still rely on manual processes and informal communication tools for their daily operations.

The strategy also highlights the need for digital commerce tools, mobile-friendly services, and data-driven business practices. These priorities are highly relevant to small enterprises that want to remain competitive in an increasingly digital marketplace. For businesses like Margaux Collections, the document provides a policy environment that supports the development of web-based ordering and inventory systems. It shows that digital transformation is not limited to large corporations; rather, it is also intended to help microbusinesses improve their operations, reach more customers, and reduce inefficiencies.

Another important point raised by the strategy is the need for locally appropriate technology solutions. The document supports the development of systems that are designed with the specific needs, resources, and digital literacy levels of Filipino entrepreneurs in mind. This is especially important for microbusinesses, which usually do not have large budgets or dedicated IT teams. The Margaux Collections Ordering System, built using PHP and MySQL, reflects this kind of practical and accessible approach. These technologies are widely used, relatively affordable, and well supported in the local development environment, making them suitable for a small business project.

Local Studies

Local studies provide the empirical basis for the proposed system because they show how Filipino businesses actually use digital tools in practice. These studies are especially valuable because they reveal the operational benefits, limitations, and adoption patterns of systems designed for small enterprises in the Philippines. In the case of Margaux Collections, the local studies reviewed here support the need for a more structured ordering platform that can improve efficiency, accuracy, and customer trust. They also demonstrate that even small-scale businesses can benefit significantly from digital solutions when these are designed according to their actual workflow, resources, and operational challenges.

Developing a Point-of-Sale Inventory Management System for Lords Grace General Merchandising

Antonio, Reyes, and Torres, 2017

The development of web-based point-of-sale and inventory management systems has become an important area of applied computing research in the Philippines, especially for small businesses that need affordable and practical digital tools. Antonio, Reyes, and Torres (2017) developed a PHP and MySQL-based inventory management system for Lords Grace General Merchandising, a small retail business in the country, as part of an undergraduate capstone project at the Polytechnic University of the Philippines. Their work demonstrates how local businesses can benefit from digital systems that help improve inventory control, sales tracking, and overall operational efficiency. It also shows that even small enterprises can successfully adopt technology when the system is designed according to their actual needs and limitations. In this sense, the study serves as a practical example of how information systems can be applied in a real business environment to solve everyday operational problems.

The system developed by Antonio et al. (2017) included several important modules that supported day-to-day retail operations. These modules covered product management, which allowed users to add, edit, and delete product records; inventory tracking, which monitored stock levels and generated low-stock alerts; sales recording, which logged transactions and produced receipts; and basic reporting, which summarized sales performance over specific periods and identified top-selling products. Together, these features created a more organized and efficient workflow compared to manual recordkeeping. Instead of relying on notebooks, memory, or scattered paper records, the business could use a centralized digital system that stored data more accurately and made it easier to access when needed. This kind of structure is especially useful in small business settings where the owner or staff may need to manage several responsibilities at the same time and cannot afford delays or mistakes in recordkeeping.

The study also showed that the adoption of a digital system helped reduce common operational problems. Manual methods often led to misplaced records, delayed updates, and inconsistencies in stock counts. By contrast, the computer-based system allowed information to be updated more quickly and consistently. This is important because inventory accuracy affects not only internal management but also customer satisfaction. When stock records are wrong, businesses may promise products that are no longer available or fail to notice items that need replenishment. Antonio et al. (2017) therefore demonstrate that a well-designed system can improve reliability and reduce preventable errors. Their findings also suggest that digital recordkeeping supports better decision-making because owners can monitor sales trends more effectively and respond to changes in product demand with greater confidence.

Another important contribution of the study is that it highlights the practicality of using widely accessible technologies such as PHP and MySQL for local business applications. These technologies are commonly used in academic and small-scale commercial system development because they are relatively affordable, flexible, and easy to maintain. For small enterprises in the Philippines, this is significant because budget limitations often prevent them from adopting expensive proprietary software. The system developed by Antonio et al. (2017) shows that a business does not need a complex or costly platform to improve its operations. What it needs is a system that is functional, user-friendly, and aligned

with its actual processes. This makes the study a useful model for the development of similar systems in other small retail businesses.

These findings are highly relevant to Margaux Collections, since the business also needs a reliable way to manage products, sales, and inventory. Like the business in the study, Margaux Collections currently faces the challenge of handling product information and transactions in a more informal way. The proposed Ordering System is designed to address these same concerns by providing a structured platform for product management, sales recording, and inventory monitoring. This makes Antonio et al. (2017) a strong local precedent for the system, because it proves that a PHP and MySQL-based solution can work effectively in a small Philippine business setting. More importantly, it demonstrates that local businesses can benefit from a formal digital system that organizes their operations, reduces errors, and supports more efficient customer service.

The relevance of this study to Margaux Collections goes beyond technical similarity. It also supports the idea that an ordering system can serve as a practical management tool that improves the overall business process. By centralizing product records and monitoring stock movements in real time, the system can help the owner avoid duplication, confusion, and manual errors. It can also make it easier to update product availability, track sales performance, and review which items are most frequently purchased. In a small fashion business that offers both brand-new and preloved products, these functions are especially important because product conditions, categories, and availability must be carefully monitored. Antonio et al. (2017) therefore provide a strong foundation for justifying the need for a similar system in Margaux Collections.

In addition, the study emphasizes that digital systems are most effective when they are designed around the actual workflow of the business. This is a crucial point for Margaux Collections because the proposed Ordering System is not intended to be a generic e-commerce platform. Instead, it is being developed to match the specific needs of the business, including managing fashion items, separating brand-new from preloved products, and tracking customer orders more systematically. The study by Antonio et al. (2017) supports this approach by showing that practical, business-specific systems are more useful than broad or overly complicated applications. As a result, the study reinforces the importance

of tailoring the system to the real operational needs of the business rather than forcing the business to adjust to an unsuitable digital tool.

Analysis of Inventory Management Systems of Small-Sized Restaurants in Quezon Province

Santos, Dela Cruz, and Lopez, 2021

Santos, Dela Cruz, and Lopez (2021), in the *Philippine Journal of Hospitality and Service Management*, examined the inventory management practices of small-sized restaurants in Quezon Province, Philippines, in order to identify effective methods and common weaknesses that could guide the development of better inventory systems. Their study used a mixed-methods approach that combined inventory audits, interviews with business owners, and a comparison of manual and digital inventory management methods. Although the setting of the study was the restaurant industry, the research offers valuable insights that extend beyond food service because it focuses on the practical challenges of tracking inventory accurately, minimizing waste, and improving daily operational efficiency. The study is especially relevant to small enterprises because it reflects the realities of businesses that often operate with limited staff, limited time, and limited resources, yet still need to maintain accurate and dependable records.

One of the central findings of the study was that manual inventory methods frequently resulted in delays, errors, and incomplete records. Small restaurant owners found it difficult to monitor their stock consistently because inventory levels changed often and updates were sometimes made only after problems had already occurred. This kind of reactive recordkeeping created confusion and made it harder to maintain accurate stock visibility. In many cases, manual methods also increased the risk of overordering, underordering, and wasting supplies, especially when the business depended on memory or handwritten logs. These challenges show that inventory management is not merely an administrative task, but a critical operational function that directly affects business continuity and service quality.

In contrast, the study found that digital inventory systems provided better control because they allowed owners to monitor stock movement more systematically and respond more quickly to changes in demand. With a digital system, updates could be made in a more organized and timely way, which improved the accuracy of records and reduced the likelihood of errors. This is important because inventory accuracy influences not only internal operations but also customer satisfaction. When a business knows what is available and what has already been used or sold, it can serve customers more efficiently and avoid disappointing them with unavailable items. For small businesses, this level of control can make a significant difference in daily performance and long-term reliability.

The study also identified three important factors that influenced successful digital inventory management adoption among small Philippine businesses: simplicity of the user interface, reliability of the system, and perceived usefulness compared with the old method. These factors are particularly important for microbusinesses because owners often have multiple responsibilities and cannot afford to spend much time learning complicated software. A system must therefore be easy to understand, dependable in actual use, and clearly better than the manual process it replaces. If a system is too complex, business owners may avoid using it consistently, which defeats the purpose of digitalization. This finding suggests that the success of an inventory system depends not only on its technical features, but also on how practical and user-friendly it is in everyday business operations.

Another significant implication of the study is that digital inventory systems are more likely to be adopted when business owners can immediately see their value. If the system helps save time, improves accuracy, and reduces stress in managing stock, then it becomes easier to justify its use. This is especially relevant in small-scale enterprises where owners often make decisions based on direct usefulness and observable outcomes. In this sense, the study demonstrates that technology adoption among small businesses is strongly shaped by practical benefits rather than abstract innovation alone. A system that clearly simplifies work and improves recordkeeping is more likely to be embraced and used consistently.

Although the restaurant industry is different from fashion retail, the core inventory management tasks are very similar. Both types of businesses must track stock

levels, record sales, identify fast-moving and slow-moving items, and determine when products need to be replenished. Both also face the challenge of ensuring that records match actual stock conditions so that business decisions can be made accurately. Because of these shared operational needs, the findings of Santos et al. (2021) are highly relevant to Margaux Collections. The proposed Ordering System should support the same basic inventory functions that the study found to be effective in restaurants, especially in terms of stock monitoring, system reliability, and ease of use.

For Margaux Collections, this means that the Ordering System should help the business owner monitor what items are available and which ones have already been sold in real time. This function is important because the business handles both brand-new and preloved fashion items, each of which must be tracked carefully to avoid confusion. Since preloved items are often unique or limited in quantity, the risk of duplication or incorrect availability is higher if inventory is not managed properly. A digital system can reduce this risk by keeping product records organized and updated. It can also help the owner identify which items are selling quickly and which are remaining in stock for longer periods, which supports better planning and decision-making.

The study by Santos et al. (2021) also supports the idea that a good inventory system should be designed with the actual needs of the business in mind. For Margaux Collections, this means that the Ordering System should not be overly complicated or filled with unnecessary features that do not contribute to daily operations. Instead, it should focus on practical functions such as product listing, stock updating, sales tracking, and order monitoring. These features can help the business operate more smoothly and professionally while reducing the burden of manual recordkeeping. In doing so, the system becomes a useful tool for both operational control and business growth.

Overall, Santos, Dela Cruz, and Lopez (2021) provide a strong local basis for the development of the Margaux Collections Ordering System. Their study shows that small businesses benefit from digital inventory systems because these improve accuracy, reduce errors, and make stock monitoring more efficient. More importantly, the study demonstrates that successful system adoption depends on simplicity, reliability, and usefulness, which are all essential qualities for a small fashion business. These insights support the need for an organized, user-friendly,

and dependable ordering system that can help Margaux Collections manage its products more effectively and respond better to customer needs.

Impact of Inventory Management on Financial Performance of MSMEs

Gaoat, Cabungcal, and Esteban, 2023

Gaoat, Cabungcal, and Esteban (2023) examined how inventory management practices affect the financial performance of MSMEs in Santiago City, Philippines. Their study used a quantitative research design and surveyed MSME owners and managers to determine how inventory management quality relates to financial performance indicators such as revenue, profit margin, and return on investment. The results provided strong evidence that inventory control is not merely an administrative task, but a business function with direct financial consequences. This makes the study highly relevant to small enterprises because it shows that the way a business handles stock can influence not only daily operations but also long-term profitability and sustainability.

The study found that businesses with better inventory practices tended to perform more effectively because they could avoid overstocking, reduce losses, and respond more accurately to customer demand. This is an important result because it shows that poor inventory control can lead to unnecessary costs, while efficient inventory management can support profitability and long-term sustainability. When stock is tracked properly, business owners are able to make better decisions about which items should be reordered, which products should be promoted, and which items are not moving quickly enough. In this way, inventory management becomes more than a background process; it becomes a strategic part of business planning and financial control.

Another important implication of the study is that inventory data should be used to guide decision-making rather than being treated as a simple recordkeeping task. The researchers emphasized that businesses can achieve better financial results when they rely on accurate and up-to-date stock information. This means that inventory records should not only show what is available but should also help owners understand sales trends, purchasing patterns, and product performance. For MSMEs, this is especially useful because they often operate with limited capital and cannot afford mistakes such as overbuying items that do

not sell or failing to restock products that customers want. Accurate inventory data therefore supports smarter planning and more efficient use of resources.

The study also suggests that good inventory management helps businesses respond more effectively to customer demand. When owners know which products are popular and which ones are slow-moving, they can adjust their sales and restocking strategies accordingly. This improves the business's ability to meet customer needs while minimizing waste and unnecessary storage costs. For small businesses, this kind of responsiveness is essential because customer preferences can change quickly, and financial margins are often tight. By maintaining control over stock levels, MSMEs can improve both customer satisfaction and operational efficiency at the same time.

For Margaux Collections, this study reinforces the importance of developing an ordering system that includes accurate stock tracking, sales summaries, and reporting features. Since the business sells both brand-new and preloved items, the owner must keep track of different item conditions, categories, and availability. A digital system can help ensure that the stock database is updated in real time, which reduces the risk of selling unavailable items or failing to notice slow-moving products. These improvements can contribute to better financial performance by helping the business owner make informed decisions about pricing, restocking, and promotion. In other words, the system can help the business not only stay organized but also operate more profitably.

The study is also relevant because it highlights the connection between operational efficiency and financial stability. When inventory is managed properly, businesses can reduce losses caused by stock mismanagement, product damage, or unnecessary purchasing. This is especially significant for a small fashion business like Margaux Collections, where every product matters and inventory decisions directly affect cash flow. A digital ordering system can support this by giving the owner a clearer overview of product movement and helping identify which items are contributing most to sales. As a result, the business can allocate attention and resources more effectively.

Another useful point from the study is that inventory management should be viewed as a continuing business process rather than a one-time task. MSMEs need systems that can adapt as their operations grow and their product lines change. For Margaux Collections, this means that the Ordering System should be flexible enough to handle new products, preloved items, and updated

customer orders without creating confusion. The ability to monitor and analyze inventory consistently can help the business remain responsive and financially stable over time.

Overall, the local studies reviewed here show that digital systems are highly valuable for small Philippine businesses. Antonio et al. (2017) demonstrate that PHP and MySQL-based systems can improve retail operations; Santos et al. (2021) show that digital inventory systems are more reliable than manual methods; and Gaoat et al. (2023) confirm that effective inventory management supports stronger financial performance. Together, these studies justify the development of the Margaux Collections Ordering System as a practical and research-based solution for a small fashion business in the Philippines. They also show that a well-designed system can strengthen both operational control and financial decision-making, which are essential for the long-term success of any MSME.

Synthesis of the Literature and Studies

The reviewed local and foreign literature and studies collectively establish that digital ordering systems are highly relevant for small fashion businesses, especially those that sell both brand-new and preloved items. Across the sources, a consistent theme emerges: consumers are more likely to purchase when products are presented clearly, pricing is transparent, and the platform is easy to use. Whether the discussion focuses on second-hand fashion, e-commerce behavior, inventory management, or sustainable supply chains, the literature repeatedly shows that organization, trust, and product clarity are essential factors in improving customer experience and business performance.

The local literature and studies show that the Philippine setting is already familiar with second-hand fashion through ukay-ukay culture and informal digital commerce through platforms such as Facebook and Messenger. However, the same sources also reveal the limitations of these informal practices, particularly in relation to inventory tracking, order monitoring, and consistency of product

presentation. This indicates that while small Filipino businesses are already digitally active, many still lack structured systems that can support efficient and reliable operations. The Margaux Collections Ordering System directly addresses this gap by providing a more formal, organized, and business-focused platform for handling products and transactions.

The foreign literature and studies further strengthen this position by showing that consumers value not only affordability but also emotional appeal, sustainability, quality, and trust. Studies on circular fashion and second-hand luxury goods demonstrate that buyers are more willing to purchase preloved items when they are described honestly, photographed well, and categorized clearly. At the same time, research on online shopping confirms that convenience must be balanced with reassurance, especially because buyers cannot physically inspect products before purchase. These findings suggest that an effective ordering system must do more than display products; it must also reduce uncertainty and build confidence.

Another important theme across the reviewed sources is sustainability. Several studies explain that preloved fashion contributes to waste reduction, garment reuse, and more responsible consumption. Sustainable supply chain literature also emphasizes the importance of accurate records, transparency, and efficient product flow. For Margaux Collections, these ideas support the development of a system that not only improves business operations but also reinforces the environmental value of buying second-hand clothing. In this way, the proposed system aligns with both commercial and social responsibility goals.

Overall, the literature and studies show that the proposed Margaux Collections Ordering System is grounded in both practical need and academic evidence. The system is justified by the realities of informal selling in the Philippines, the growing acceptance of second-hand fashion, and the importance of digital tools in improving efficiency and customer trust. By integrating inventory management, product presentation, order processing, and clear communication, the system responds to the issues identified in the reviewed literature and studies while supporting the long-term growth of the business.

Research Gap

Although the reviewed studies provide valuable insights, they also reveal a gap that the present study aims to address. Many foreign studies focus on second-hand luxury consumption, circular fashion, or broad e-commerce behavior, while local studies often examine general microbusiness practices or inventory systems in other industries. There is still limited research that specifically examines a web-based ordering system designed for a small Philippine fashion business that sells both new and preloved items. This gap is important because the operational needs of such a business are unique: it requires product categorization, condition labeling, customer communication, and inventory tracking in one organized platform.

The present study addresses this gap by developing a tailored ordering system for Margaux Collections. Unlike general e-commerce platforms, the proposed system is designed around the actual workflow of the business and the nature of its products. It aims to improve the management of brand-new and preloved fashion items, strengthen customer trust through clearer product presentation, and support better decision-making through organized records and sales monitoring. In doing so, the study contributes to the growing body of research on digital transformation among Filipino microbusinesses.

Opening Paragraph

This chapter presents the related literature and studies that serve as the foundation of the Margaux Collections Ordering System. It gathers discussions from local and foreign sources to explain the concepts, theories, findings, and practical applications that are relevant to the development of the proposed system. The review focuses on second-hand fashion, digital ordering, consumer behavior, inventory management, sustainable business practices, and the use of information systems in small enterprises. By examining these studies and literature, the chapter establishes the academic and practical justification for developing a more organized and efficient platform for managing the products and transactions of Margaux Collections.

Closing Paragraph

The research shows that a web-based system for ordering is just what a small fashion business needs, especially one that sells new and used items. It's all about showing products clearly, keeping track of inventory, building trust with customers, and making business processes run smoothly. People are really into second-hand fashion now, and it's good for the planet too. Plus, more and more businesses are using digital platforms, so it makes sense for fashion to do the same. The Margaux Collections Ordering System is more than just a technical fix - it's a way to meet the needs of the business and what customers expect these days. The information gathered here gives a solid foundation for creating the proposed system, which could make fashion retail more organized, reliable, and sustainable. It's pretty clear that customers want to be able to trust the businesses they buy from, and they want the process to be easy and efficient. A good web-based ordering system can help with that. And with more people shopping online, it's a great way for small fashion businesses to reach more customers and grow. The key is to make sure the system is user-friendly, and that it helps the business run more smoothly behind the scenes. By developing a system like this, small fashion businesses can stay competitive and appeal to customers who care about sustainability. It's a win-win, really - the business gets to operate more efficiently, and the customer gets a better experience. And with the growing demand for second-hand fashion, it's an exciting time for businesses that want to get in on this trend. Overall, the research suggests that a web-based ordering system is a great idea for small fashion businesses, and it could be a big part of making the fashion industry more sustainable. By investing in this kind of technology, businesses can set themselves up for success and help create a more environmentally-friendly fashion scene.

References

Foreign Literature

- Gansky, L. (2010). *The mesh: Why the future of business is sharing*. Penguin.
- Acquisti, A., Taylor, C., & Wagman, L. (2016). The economics of privacy. *Journal of Economic Literature*, 54(2), 442–492.
- Vanguard sources on sustainable fashion and business models:
 - Vänskä, A., Rauti, S., Heino, T., Carlsson, R., Mickelsson, S., & Särmäkari, N. (2024). Fair data is the new black: Online shopping, data leaks, and broadening the understanding of sustainable fashion. *Fashion Theory*, 28(3), 305–333.
 - Cline, E. (2013). *Overdressed: The shockingly high cost of cheap fashion*. Portfolio.
 - Vladimirova, K. (2021). Consumption corridors in fashion: Deliberations on upper consumption limits in minimalist fashion challenges. *Sustainability: Science, Practice and Policy*, 17(1), 102–116.
 - Brand, S., Jacobs, B., & Taljaard-Swart, H. (2023). I rent, swap or buy second-hand – comparing antecedents for online collaborative clothing consumption models. *International Journal of Fashion Design, Technology and Education*, 16(3), 275–287.
 - Abimbola, O. (2012). The international trade in secondhand clothing: Managing information asymmetry between West African and British traders. *TEXTILE*, 10(2), 184–199.

- Ayres, J. (2019). The work of shopping: Resellers and the informal economy at the goodwill bins. *Business History*, 61(1), 122–154.
- Frazer, G. (2008). Used-clothing donations and apparel production in Africa. *The Economic Journal*, 118(532), 1764–1784.
- Amatulli, C., Guido, G., Mileti, A., Tomacelli, C., Prete, M. I., & Longo, A. E. (2016). “Mix-and-match” fashion trend and luxury brand recognition: An empirical test using eye-tracking. *Fashion Theory*, 20(3), 341–362.
- Skjulstad, S., & Hvidsten, A. (2025). Backstage matters: Designer’s invisible media work in independent fashion practice. *Fashion Theory*, 29(2), 171–191.
- Svensson, G. (2007). Aspects of sustainable supply chain management (SSCM): Conceptual framework and empirical example. *Supply Chain Management: An International Journal*, 12(4), 262–266.
- Todeschini, B. V., Cortimiglia, M. N., Callegaro-de-Menezes, D., & Ghezzi, A. (2017). Innovative and sustainable business models in the fashion industry: Entrepreneurial drivers, opportunities, and challenges. *Business Horizons*, 60(6), 759–770.

Foreign Studies

These are empirical studies with methods, findings, or data analysis:

- Araújo, R. G. (2021). The determinants and consequences of customer experience in e-commerce pre-loved luxury fashion. Master’s thesis, Universidade do Porto, Portugal.
- Armstrong, C. M. J., & Park, H. (2020). Online clothing resale: A practice theory approach to evaluate sustainable consumption gains. *Journal of Sustainability Research*, 2(2), e200017.

- Clube, R. K., & Tennant, M. (2020). Exploring garment rental as a sustainable business model in the fashion industry: Does contamination impact the consumption experience? *Journal of Consumer Behaviour*, 19(4), 359–370.
- Cuc, S., & Vidovic, M. (2014). Environmental sustainability through clothing recycling. *Operations and Supply Chain Management: An International Journal*, 4(2), 108–115.
- Gaoat, E. C., Cabungcal, M. R., & Esteban, M. S. (2023). Impact of inventory management on financial performance of MSMEs in Santiago City. *Asian Journal of Economics, Business, and Industry*, 11(3), 18–29.
- Gopalakrishnan, S., & Matthews, D. (2018). Collaborative consumption: A business model analysis of second-hand fashion. *Journal of Fashion Marketing and Management: An International Journal*, 22(3), 354–368.
- Kim, I., Jung, H. J., & Lee, Y. (2021). Consumers' value and risk perceptions of circular fashion: Comparison between secondhand, upcycled, and recycled clothing. *Sustainability*, 13(3), 1208.
- McNeill, L., & Moore, R. (2015). Sustainable fashion consumption and the fast fashion conundrum: Fashionable consumers and attitudes to sustainability in clothing choice. *International Journal of Consumer Studies*, 39(3), 212–222.
- Neto, J. Q. F., Bloemhof, J., & Corbett, C. (2016). Market prices of remanufactured, used and new items: Evidence from eBay. *International Journal of Production Economics*, 171, 371–380.
- Santos, L. M., Dela Cruz, R. J., & Lopez, M. A. (2021). Analysis of inventory management system of selected small-sized restaurants in Quezon Province: Basis for an inventory system manual. *Philippine Journal of Hospitality and Service Management*, 9(2), 15–28.

- Sarkar, R., & Das, S. (2017). Online shopping vs offline shopping: A comparative study. *International Journal of Scientific Research in Science and Technology*, 3(1), 424–431.
- Tu, J. C., Hsu, C. F., & Creativani, K. (2022). A study on the effects of consumers' perception and purchasing behavior for second-hand luxury goods by perceived value. *Sustainability*, 14(16), 10397.
- Turunen, L. L. M., Cervellon, M. C., & Carey, L. D. (2020). Selling second-hand luxury: Empowerment and enactment of social roles. *Journal of Business Research*, 116, 474–481.
- Yan, R. N., Bae, S. Y., & Xu, H. (2015). Second-hand clothing shopping among college students: The role of psychographic characteristics. *Young Consumers*, 16(1), 85–98.

Local Sources

These are Philippine-based and can be placed under local literature or local studies depending on your Chapter 2 structure:

- Antonio, J. R., Reyes, M. L., & Torres, C. R. (2017). Developing a point-of-sale inventory management system for Lord's Grace General Merchandising using PHP and MySQL. Unpublished undergraduate capstone, Polytechnic University of the Philippines.
- Binuluan, A. C. E., Gonayon, S. J. D., Mabborang, A. M. R., Panaligan, M. R. C., & Ramos, R. (2024). Social media utilization of microbusinesses in a component city in Northern Philippines. *International Journal of Marketing and Digital Creative*, 2(2), 1–12.
- Department of Information and Communications Technology (DICT). (2022). *Philippine Digital Transformation Strategy 2022*. Department of Information and Communications Technology, Republic of the Philippines.

- Isla, V. L. (2013). Investigating second-hand fashion trade and consumption in the Philippines: Expanding existing discourses. *Journal of Consumer Culture*, 13(3), 221–240.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study focuses on the design, development, testing, and evaluation of the Margaux Collections Ordering System. The system serves as a web-based platform for managing orders and inventory for a small fashion business that deals in both branded-new bikinis, co-ords, and accessories and preloved pants, skirts, and shorts. Given that the project involves creating a functional software application aimed at solving real-world operational problems, the study adopts a developmental (systems development) research design.

A developmental research design is appropriate because:

- It is used when the primary objective is to develop and refine a system, not just to describe or correlate variables.
- It emphasizes iterative design, continuous testing, and feedback from real users.
- It supports the evaluation of the system's performance, usability, reliability, and security in a practical business context.

The Margaux Collections Ordering System is developed and systematically evaluated to determine whether it:

- Accurately records and processes orders,
- Effectively manages both new and preloved inventory,
- Provides a user-friendly interface for customers and the owner, and
- Integrates smoothly into the existing Facebook-based marketing and selling workflow.

By following a developmental design, the study ensures that the system is practically useful, well-documented, and supported by empirical evidence from user testing and performance evaluation.

3.2 System Development Methodology

The Margaux Collections Ordering System is developed using the Agile Software Development methodology, specifically an iterative and incremental approach suitable for small-scale web applications. Agile methodology is chosen because it:

- Allows the development team to work in short, focused sprints.
- Encourages continuous feedback from the business owner and potential users.
- Enables the team to adapt to changing requirements during development.
- Supports regular testing, code review, and incremental improvement of features.

3.2.1 Requirements Analysis

- Conducting interviews with the business owner to understand current business processes, pain points, and expectations for the system.
- Observing the existing manual ordering and inventory process to identify inefficiencies in order processing, product tracking, and record-keeping.
- Collecting data from existing product lists, pricing sheets, and order records to define the basic data structure.
- Designing user stories that describe how both the customer and the owner will interact with the system (e.g., “As a customer, I want to view preloved items separately from new items”).
- Translating user stories into technical requirements such as:
 - Must support two main product categories: new items and preloved items.
 - Must allow adding, updating, and deleting products and orders.
 - Must provide a dashboard for viewing daily sales and inventory status.

The result of this phase is a requirements specification document that serves as the foundation for the system’s design and development.

3.2.2 System Design

- Designing the overall system architecture (front end, back end, and database).
- Creating wireframes and low-fidelity mockups of the user interface for both the customer view and the admin panel.
- Designing the database schema (tables, relationships, and constraints).
- Preparing data flow diagrams (DFD) and entity relationship diagrams (ERD) to show how data move across the system.

3.2.3 Development

- Implementing the front-end interface (HTML, CSS, JavaScript) for product browsing, shopping cart, checkout, and order history.
- Developing the back-end logic (PHP) for user authentication, order processing, inventory updates, and analytics.

- Building the database (MySQL) and connecting it securely to the PHP scripts.
- Conducting code reviews and peer testing within the team to catch errors early.
- Integrating basic analytics and reporting features (e.g., sales summary, best-selling items list).

3.2.4 Testing

- Unit testing of individual PHP functions and database queries.
- Integration testing of the entire system to ensure that front-end, back-end, and database layers work together.
- Security testing to identify vulnerabilities such as injection attacks or unauthorized access.
- User testing with the business owner and selected customers to evaluate usability and accuracy of order processing.

3.2.5 Deployment (Prototype)

After sufficient testing, a working prototype of the Margaux Collections Ordering System is deployed in a local or test server environment and presented to the business owner for evaluation. The owner can:

- Place test orders
- Update inventory, and
- Use the analytics dashboard to review performance.

Based on the owner's feedback, the team may perform minor refinements before finalizing the system. This iterative deployment approach ensures that the system aligns closely with real business needs.

3.3 System Architecture

The Margaux Collections Ordering System follows a three-layer architecture to ensure separation of concerns, maintainability, and scalability. The three layers are:

1. Presentation Layer (Front End)
2. Application Layer (Back End / Business Logic)
3. Data Layer (Database)

Each layer has a distinct role and responsibility, allowing the system to be developed, tested, and maintained efficiently.

3.3.1 Presentation Layer

The Presentation Layer is the front-end face of the system that users interact with. It is implemented using:

- HTML5 – for structuring the web pages.
- CSS3 – for styling and layout.
- **JavaScript** – for dynamic behavior such as updating the shopping cart and showing real-time validation.

This layer provides:

- **Customer Interface**
 - Product browsing page showing **new items (bikinis, co-ords, accessories)** and **preloved items (pants, skirts, shorts)** in separate sections or filters.
 - **Shopping cart** page where users can add, remove, or update quantities of items.
 - **Checkout page** for entering or confirming delivery details, selecting payment method, and confirming the order.
 - **Order history page** where customers can view past orders and statuses (e.g., “pending,” “for delivery”).
- **Admin Interface**
 - **Dashboard** showing total sales, number of orders, and inventory status.

- **Products management page** where the owner can add, update, or delete products, including setting **product type (new or preloved)**.
- **Orders management page** for viewing and updating order status.
- **User management page** for viewing and managing customer data (if applicable).

The interface is designed with **responsive CSS techniques** so that it works well on **desktops, tablets, and mobile phones**, which is important since many customers may access it from their mobile devices via Facebook links.

3.3.2 Application Layer

The **Application Layer** represents the **back-end logic** of the system. It is developed using **PHP** and acts as the middle layer between the **front-end** and the **database**. This layer is responsible for:

- **User authentication and authorization** (if user accounts are implemented).
- **Order processing**, including assigning order numbers, calculating total amounts, and updating inventory.
- **Inventory management**, ensuring that stock quantities are correctly updated when orders are placed or canceled.
- **Status updates**, such as changing an order from “pending” to “for delivery” or “completed.”
- **Security mechanisms**, including:
 - Protection against **SQL injection** through **prepared statements**.
 - **Session management** with timeout and secure session handling.
 - **Input validation** for all user-entered data.
 - **Rate limiting** on login or critical actions (if applicable).

The PHP-based application layer also enforces **role-based access control**:

- **Customers** – can view products, add items to the cart, check out, and view order history.
- **Owner/Administrator** – can manage products, orders, and users, and access the analytics dashboard.

This separation ensures that sensitive operations are restricted to authorized users only.

3.3.3 Data Layer

The Data Layer handles:

- Storing user information (if accounts are used).
- Storing product information (name, price, description, type, stock, image path).
- Storing order transactions (customer details, order items, total amount, payment status).
- Storing order-item details (which products are included in each order and their quantities and prices).

The database is designed with proper normalization to avoid redundancy and ensure data integrity. Indexing is applied to frequently queried fields (e.g., order IDs, product IDs) to improve query performance. These design choices help guarantee the reliability, efficiency, and security of data operations throughout the Margaux Collections Ordering System.

3.4 System Features and Modules

The **Margaux Collections Ordering System** is composed of several **core modules** that work together to provide a complete ordering and management platform. Each module addresses a specific aspect of the business workflow, from **user authentication** to sales analytics.

3.4.1 Authentication Module

The **Authentication Module** (if user accounts are implemented) handles:

- **User registration:** Allowing customers to create accounts with basic information (name, email, contact number).
- **Login and logout:** Verifying user credentials and managing active sessions.
- **Session management:** Keeping users logged in securely, implementing session timeouts, and preventing session hijacking.

Even if the system is designed to be **brochure-style without accounts** (using only **Facebook for inquiries**), the Authentication Module can still manage **admin login** for the owner, ensuring that only the authorized person can access the **admin panel**.

3.4.2 Product Management Module

The Product Management Module allows the owner/administrator to:

- Add new products (both new and preloved items).
- Update product details (price, description, stock, images, product type).
- Delete or deactivate products when they are no longer available.

Key features include:

- Support for two main categories:
 - New items (bikinis, co-ords, accessories).
 - Preloved items (pants, skirts, shorts).
- Option to tag products (e.g., “preloved, good condition,” “new, limited stock”).
- Real-time stock counter that updates when products are sold.

3.4.3 Shopping Cart Module

The **Shopping Cart Module** creates and manages a **virtual shopping cart** for each session or user. It:

- Allows customers to **browse products and add items** to the cart.
- **Verifies product availability** (i.e., stock quantity) before adding items.
- Supports **updating item quantities** (increase, decrease, or remove).
- Calculates **subtotals and totals** based on product prices and quantities.
- Preserves the cart contents during the checkout process even if the user navigates away temporarily.

The Shopping Cart Module is a core feature that makes the ordering process **convenient and intuitive** for customers.

3.4.4 Order Processing Module

The **Order Processing Module** manages the **entire order lifecycle**, from placement to completion. It:

- Assigns a **unique order number** to each new order.
- Captures **customer information** (name, contact number, delivery address if applicable).
- Records **product selections** (including quantities and prices at the time of purchase).
- Calculates the **total order amount** and applies any discounts (if implemented).
- Supports **various payment methods**, such as:
 - Cash on pickup / delivery.
 - Online payment via GCash, PayMaya, or other platforms (if implemented).
- Updates **product stock levels** in the database after the order is confirmed.
- Stores **order status** (e.g., “pending,” “for delivery,” “completed”) that can be viewed by both the owner and the customer.

3.4.5 Preloved-Specific Management Module

Because **Margaux Collections** heavily relies on **preloved fashion**, the system includes a **Preloved-Specific Management Module** that:

- Allows the owner to **tag items explicitly as “preloved”**.
- Includes **condition fields** (e.g., “excellent,” “good,” “fair”) so customers know the state of the item.
- May include **care instructions** or **style notes** that can be displayed on the product page.

This module differentiates the **preloved experience** from the **new items**, helping customers make informed decisions.

3.4.6 User Management Module

The **User Management Module** (if user accounts are created) enables the administrator to:

- **View** a list of registered customers.
- **Search** and **filter** users by name, contact number, or order history.
- **Edit** user information.
- **Delete or deactivate** user accounts.

This module supports **data management** and **customer relationship management** in a structured way.

3.4.7 Analytics and Sales Reporting Module

The **Analytics and Sales Reporting Module** provides the owner with a **dashboard** that displays key performance indicators such as:

- Total **sales revenue** over a selected period.
- Number of **orders placed** and **items sold**.
- **Best-selling items** (both new and preloved).
- Inventory status (items that are low-stock or out-of-stock).

This module helps the owner **monitor business performance**, **identify trends**, and **make informed decisions** about restocking and pricing.

3.4.8 Summary of Modules

Together, the following modules form the **integrated Margaux Collections Ordering System**:

- Authentication Module
- Product Management Module
- Shopping Cart Module
- Order Processing Module
- Preloved-Specific Management Module
- User Management Module
- Analytics and Sales Reporting Module

These modules ensure that the system fully supports **order placement**, **inventory tracking**, and **business analytics** for a small fashion retailer focused on **both new and preloved fashion**.

3.5 Database Design

The **Margaux Collections Ordering System** database is built on **MySQL** and follows a **relational model** with proper normalization and relationships. The main tables are designed to support core business operations.

3.5.1 Main Database Tables

The system database consists of the following primary tables:

- **Users**

Stores user account information such as:

- User ID (primary key)
- Name
- Email
- Hashed password
- Contact number
- Delivery address (if applicable)
- User role (e.g., customer, admin)
- Date created

- **Products**

Stores product catalog information such as:

- Product ID (primary key)
- Product name
- Description
- Price
- Product type (new or preloved)
- Condition (for preloved items)
- Image file path
- Stock quantity
- Date added

- **Orders**

Stores complete order transaction records such as:

- Order ID (primary key)
- User ID (foreign key)
- Customer name
- Contact number
- Delivery address

- Order method (pickup/delivery)
- Payment method (cash, online, etc.)
- Payment status (paid/unpaid)
- Order status (pending/for delivery/completed/cancelled)
- Total amount
- Order date
- **Order_Items**
Stores individual product line items within each order, including:
 - Item ID (primary key)
 - Order ID (foreign key)
 - Product ID (foreign key)
 - Quantity ordered
 - Unit price at the time of purchase
- **User_Payment_Accounts**
Stores saved payment account details (if saving payment methods is implemented), including:
 - Account ID (primary key)
 - User ID (foreign key)
 - Account type (e.g., GCash, PayMaya)
 - Account name
 - Account number
 - Default flag

These tables collectively form the **relational database structure** of the **Margaux Collections Ordering System**, ensuring that data are stored accurately and can be queried efficiently.

3.5.2 Data Integrity and Optimization

The database design includes:

- **Primary keys and foreign keys** to enforce relational integrity.
- **Appropriate normalization** (e.g., avoiding storing repeated product information in the Orders table) to reduce redundancy.
- **Indexes on frequently searched fields** (e.g., Product ID, Order ID) to speed up queries.

These design choices ensure **data consistency**, **efficient querying**, and **scalability** for future business growth.

3.6 Development Tools and Technologies

The **Margaux Collections Ordering System** is developed using the following tools and technologies:

- **PHP** – The server-side programming language used for implementing all back-end logic, including order processing, inventory updates, and authentication.
- **MySQL** – The relational database management system used for storing and retrieving all system data.
- **HTML5** – Used for structuring the web pages and defining the layout of the user interface.
- **CSS3** – Used for styling the interface, including responsive design for mobile and desktop views.
- **JavaScript** – Used for adding interactivity, such as updating the shopping cart and validating form inputs.
- **Chart.js** – A JavaScript library used to generate graphs and charts for the analytics dashboard (e.g., sales trend charts, product performance charts).
- **XAMPP** – A local development environment that integrates **Apache**, **PHP**, and **MySQL**, allowing the team to test and prototype the system on a local machine.
- **Visual Studio Code** – The primary code editor used for writing, debugging, and organizing PHP, HTML, CSS, and JavaScript files.
- **Git and GitHub** – A version control system and collaborative platform used for storing the source code, tracking changes, and supporting team coordination.

3.7 Data Gathering Techniques

During the **requirements analysis phase**, the development team employs several **data gathering techniques** to understand the business context and user needs:

- **Interviews**
 - Structured and semi-structured interviews are conducted with the **Margaux Collections owner** to gather information about current workflows, ordering methods, inventory tracking, and pain points.
 - Questions focus on:
 - How orders are currently received (via Facebook Messenger, in-person, etc.).
 - How inventory is tracked (paper list, spreadsheet, mental notes).
 - What features the owner finds most useful in a future system.
- **Observation**
 - The team observes the **actual ordering and packing process** to identify inefficiencies such as:
 - Unorganized product lists.
 - Manual order entry errors.
 - Difficulty in tracking preloved items separately.
- **Document Analysis**
 - Existing **product lists, price sheets, and order records** are reviewed to understand:
 - How products are categorized.
 - How prices are structured.
 - How orders are documented.
- **Surveys (Optional)**
 - If feasible, short surveys may be distributed to **existing customers** to collect:
 - Preferences regarding online ordering.
 - Desired features (e.g., separate preloved section, clear product images).

3.8 Testing Procedures

The system undergoes several testing procedures to ensure that it is functional, secure, and user-friendly. Testing is important because it helps identify errors

before the system is fully used in the business environment. A systematic testing process increases confidence in the final product.

The **Margaux Collections Ordering System** undergoes a **comprehensive testing process** throughout development to ensure that it is **functional, secure, and user-friendly**. The testing procedures are organized into four main types:

3.8.1 Unit Testing

Unit testing checks individual functions such as order creation, inventory updating, and validation routines. Each function is tested with different inputs to verify that it behaves correctly. Database queries are also checked to ensure that data are inserted, updated, and retrieved properly. This stage confirms that each component works before being combined with other parts.

- Individual **PHP functions** (e.g., order-creation function, stock-update function) are tested with different inputs to verify that they return the expected outputs.
- **Database queries** are tested to ensure they retrieve and update data correctly.
- Simple **front-end validation functions** (e.g., checking if required fields are filled) are also verified.

Unit testing confirms that each component performs as expected in isolation before integration.

3.8.2 Integration Testing

Integration testing checks whether the front-end, back-end, and database work together correctly. For example, when a customer adds an item to the cart or places an order, the system should update the relevant records without breaking. This stage is necessary because even if individual functions work, problems can still happen when they are connected. Integration testing ensures that the entire workflow operates smoothly.

- The interaction between **front-end, back-end, and database** is tested. For example:

- Ensuring that when a customer clicks “Add to Cart,” the product is correctly stored in the session or database.
- Verifying that when an order is placed, the **order record and order-item records** are created and **stock levels are updated**.
- Data flow is checked so that updates in the **admin panel** (e.g., price changes) are correctly reflected in the **customer view**.

3.8.3 Security Testing

Security testing checks for common vulnerabilities such as SQL injection, cross-site scripting, and session-related weaknesses. The system is also tested to ensure that access controls prevent unauthorized users from entering the admin panel. These tests are important because the system stores personal and transactional information. A secure system helps protect both customer data and business records.

- The system is evaluated for common vulnerabilities such as:
 - **SQL injection** (mitigated by using prepared statements).
 - **Cross-site scripting (XSS)** (mitigated by escaping user inputs).
 - **Cross-site request forgery (CSRF)** (mitigated by using tokens where necessary).
 - **Session hijacking** (mitigated by proper session handling and secure transport).
- Access controls are tested to ensure:
 - Customers cannot access the admin panel.
 - Admins cannot bypass authentication.

Security testing helps protect **user data** and the **integrity of the system**.

3.8.4 User Acceptance Testing (UAT)

After the prototype is deployed, the business owner and a small group of customers test the system through real or simulated tasks. They may place orders, update products, or review the dashboard. Their feedback is collected to evaluate usability, design, and accuracy. User Acceptance Testing ensures that the system works not only in theory, but also in actual use.

- After the prototype is deployed, the **Margaux Collections owner** and a small group of **selected customers** test the system by:
 - Placing real or simulated orders.
 - Updating products and inventory.
 - Using the analytics dashboard.
- Feedback is collected on:
 - Ease of use.
 - Accuracy of order and inventory processing.
 - Appearance and layout of the interface.

User Acceptance Testing ensures that the system meets **operational requirements** and is **practically usable** in the real business setting.

3.9 Evaluation Criteria

The system is evaluated using five main criteria: functionality, usability, security, performance, and reliability. Functionality checks whether the required features work correctly, such as order recording and stock updating. Usability measures how easy the interface is to learn and use, especially on mobile devices. Security checks whether data is properly protected and access is restricted.

Performance evaluates how quickly the system loads and responds under normal use. Reliability checks whether the system consistently records orders and maintains data integrity. These criteria give a clear framework for judging whether the system is ready for real business use. They also help identify areas for further improvement if needed.

The **Margaux Collections Ordering System** is systematically evaluated using a **standardized assessment framework** based on the following criteria:

3.9.1 Functionality

- The system must **perform all required functions accurately**.
- This includes:
 - Correct recording of orders.
 - Proper updating of inventory.

- Accurate calculation of totals.
- Clear display of order history and status.

3.9.2 Usability

- The interface must be **intuitive and easy to use** for both customers and the owner.
- The design should:
 - Minimize the learning curve.
 - Provide clear navigation and visual feedback.
 - Be accessible on mobile devices.

3.9.3 Security

- The system must **protect user data** and prevent unauthorized access.
- Data must be:
 - Properly hashed and stored.
 - Transmitted securely.
 - Accessible only to authorized users.

3.9.4 Performance

- The system must respond **quickly and smoothly** under normal usage.
- It should:
 - Load pages within a reasonable time.
 - Handle concurrent users without significant lag.

3.9.5 Reliability

- The system must **process orders correctly** and maintain **data integrity** across all operations.
- Orders and inventory changes must be:
 - Consistently recorded.
 - Undoable or reversible only through formal processes (e.g., cancellations).

3.10 Data Flow Diagram (DFD — Level 0)

The Level 0 Data Flow Diagram shows how customers and administrators interact with the system and how data move to the MySQL database. Customers submit registration data, login credentials, order details, and checkout confirmations. The system processes these inputs by validating product availability, creating order records, and updating inventory. It then stores the information in the database and returns confirmations, product listings, or order status updates.

Administrators submit product updates, inventory adjustments, and order status changes. The system processes these changes and updates the database in real time. The DFD helps explain the overall structure of the system and shows how data move between users, the application, and storage. It is useful because it makes the system's workflow easier to understand.

Main Data Flows

- **Customer → Margaux Collections Ordering System → MySQL Database**
 - Customers submit:
 - Registration details (if accounts are used).
 - Login credentials.
 - Order information (product selections, contact details, payment method).
 - Checkout confirmation.
 - The system processes:
 - Authentication.
 - Validation of product availability.
 - Creation of order records.
 - Updating inventory.
 - The system stores transaction data in the **MySQL database**.
 - The system retrieves and displays:
 - Product listings.
 - Order confirmations.

- Order status and history.
- **Administrator → Margaux Collections Ordering System → MySQL Database**
 - The administrator submits:
 - Product updates (add, edit, delete).
 - Order status changes (e.g., pending → for delivery).
 - Inventory adjustments.
 - The system processes these inputs and updates the **database in real time**.
 - Updates are reflected immediately in the **customer view** and **analytics dashboard**.

The Level 0 DFD provides a **high-level view** of how data move between external actors and the system, making it easier to understand the overall structure of the information system.

3.11 Entity Relationship Diagram (ERD)

The ERD defines the main entities in the database: User, Product, Order, OrderItem, and UserPaymentAccount. Each entity has its own attributes and relationships. A user can place many orders, an order can contain many order items, and each product can appear in many order items. If payment accounts are included, a user can also have multiple saved payment accounts.

The ERD is important because it shows how the database is logically organized. It helps ensure that the tables are connected properly and that information can be retrieved accurately. A clear ERD also supports good database design and easier maintenance.

The **Entity Relationship Diagram (ERD)** defines the **core entities, their attributes, and relationships** within the **Margaux Collections Ordering System** database.

Entities and Key Attributes

- **User**

- user_id (PK)
- name
- email
- password (hashed)
- contact_number
- address
- role (customer, admin)
- created_at

- **Product**

- product_id (PK)
- product_name
- description
- price
- product_type (new, preloved)
- condition (for preloved items)
- image_path
- stock_quantity
- added_at

- **Order**

- order_id (PK)
- user_id (FK)
- customer_name
- contact_number
- address
- order_method (pickup, delivery)
- payment_method
- payment_status (paid, unpaid)
- order_status (pending, for delivery, completed, cancelled)
- total_amount
- order_date

- **OrderItem**

- item_id (PK)
- order_id (FK)
- product_id (FK)
- quantity
- unit_price

- **UserPaymentAccount**
 - account_id (PK)
 - user_id (FK)
 - account_type (e.g., GCash, PayMaya)
 - account_name
 - account_number
 - is_default

Relationships

- **Users** can place multiple **Orders** (1:N).
- **Order** can contain multiple **OrderItems** (1:N).
- **Products** can appear in multiple **OrderItems** (1:N).
- **Users** can have multiple **UserPaymentAccounts** (1:N).

3.12 Use Case Diagram

The Use Case Diagram presents the two main actors in the system: Customer and Administrator. Customers can register, log in, browse products, add items to the cart, view the cart, proceed to checkout, place orders, view order history, update profiles, and save payment methods. These use cases reflect the main customer actions in the ordering process.

Administrators can log in to the admin panel, manage products, manage orders, manage users, and view analytics. These actions represent the business management side of the system. The use case diagram helps show the functional scope of the system and clarifies what each actor is allowed to do. It is a useful visual summary of how the Margaux Collections Ordering System is expected to operate.

The **Margaux Collections Ordering System** is represented by a **Use Case Diagram** with two primary actors:

- **Customer**
- **Administrator**

Customer Use Cases

- **Register** – Register for an account (if the system supports accounts).
- **Login** – Log in to the system to view personal information and order history.
- **Browse Products** – View the product catalog, including filters for **new items** and **preloved items**.
- **Add to Cart** – Add selected products to the shopping cart.
- **View Cart** – See the items in the cart and update quantities.
- **Proceed to Checkout** – Confirm the order, enter delivery details, and choose a payment method.
- **Place Order** – Submit the order and receive an order confirmation and reference number.
- **View Order History** – Check the status of past orders.
- **Update Profile** – Update personal information and contact details.
- **Save Payment Methods** – Save payment account details (e.g., GCash, PayMaya) for future use.

Administrator Use Cases

- **Login to Admin Panel** – Log in securely to the admin dashboard.
- **Manage Products** – Add, edit, or delete products, including setting **product type** (new or preloved).
- **Manage Orders** – View all orders and update their status (e.g., mark as “for delivery” or “completed”).
- **Manage Users** – View and manage customer accounts (if accounts are used).
- **View Analytics** – Access the **analytics dashboard** to see sales reports, top-selling items, and inventory status.